

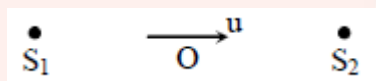
JEE Mains 12th April, 2019 Evening

Physics

1. Two sources of sound S_1 and S_2 produce sound waves of same frequency 660 Hz. A listener is moving source S_1 towards S_2 with a constant speed u m/s and he hears 10 beats /s. The velocity of sound is 330 m/s. Then, u equals:

- A. 2.5 m/s B. 10.0 m/s
C. 15.0 m/s D. 5.5 m/s

Ans: A



When observer goes away from source S_1 towards S_2 , then due to Doppler's effect so apparent frequency is given by

$v_1 = \frac{(v - v_0)}{v} v$ where v = speed of sound, v_0 = speed of observer

$$= \frac{330 - u}{330} \times 660$$

$$v_1 = 2 \times 330 - 2u \dots (1)$$

when observer goes towards source S_2 then apparent frequency is given by

$$v_2 = \frac{(v + v_0)}{v} v$$

$$= \left(\frac{330 + u}{330} \right) \times 660$$

$$v_2 = 2 \times 330 + 2u \dots (2)$$

According to question

$$v_2 - v_1 = 10$$

$$2 \times 330 + 2u - 2 \times 330 + 2u = 10$$

$$4u = 10$$

$$u = 2.5 \text{ m/s}$$

2. The electron in a hydrogen atom first jumps from the third excited state to the second excited state and subsequently to the first excited state. The ratio of the respective wavelengths, λ_1 / λ_2 , of the photons emitted in this process is :

A. $27 / 5$

B. $20 / 7$

C. $7 / 5$

D. $9 / 7$

Ans: B

Wavelength λ of emitted photon as an electron transits from an initial energy level n_i to some final energy level n_f is given by Balmer's formula,

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Where, R = Rydberg constant.

In transition from $n = 4$ to $n = 3$, we have

$$\begin{aligned} \frac{1}{\lambda_1} &= R \left(\frac{1}{3^2} - \frac{1}{4^2} \right) \\ &= R \left(\frac{7}{9 \times 16} \right) \dots (i) \end{aligned}$$

In transition from $n = 3$ to $n = 2$, we have

$$\begin{aligned} \frac{1}{\lambda_2} &= R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \\ &= R \left(\frac{5}{9 \times 4} \right) \dots (ii) \end{aligned}$$

So, from Eqs. (i) and (ii), the ratio of $\frac{\lambda_2}{\lambda_1}$ is

$$\frac{\lambda_2}{\lambda_1} = \frac{\left(\frac{7}{9 \times 16}\right)}{\left(\frac{5}{9 \times 4}\right)}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{7}{9 \times 16} \times \frac{9 \times 4}{5} = \frac{\lambda_2}{\lambda_1} = \frac{7}{20}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{20}{7}$$

3. A small speaker delivers 2 W of audio output. At what distance from the speaker will one detect 120 dB intensity sound ? [Given reference intensity of sound as 10^{-12} W / m^2]

A. 10 cm B. 30 cm
C. 20 cm D. 40 cm

Ans: D

Loudness of sound in decibel is given by

$$L = 10 \log \frac{I}{I_0}$$

Where, I = intensity of sound ($\because L = 120 \text{ dB}$) I_0 = reference intensity ($= 10^{-12} \text{ w/m}^2$)

$$120 = 10 \log \frac{I}{I_0}$$

Taking antilog, we get

$$12 = \log_{10} \frac{I}{10^{-12}}$$

$$\frac{I}{10^{-12}} = 10^{12} \quad \dots(i)$$

Now, intensity

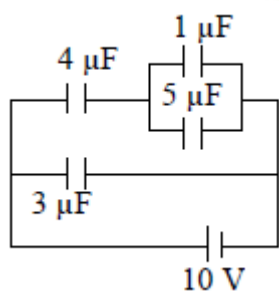
$$I = \frac{P}{4\pi r^2}$$

Here, p = 2W, I = 1

$$\text{so, } r^2 = \frac{2}{4\pi \times 1}$$

$$r = \sqrt{\frac{2}{4\pi}} \text{ m} = 100 \times \sqrt{\frac{1}{2\pi}} = 100 \times 0.399 \approx 39.9 \approx 40 \text{ cm}$$

4. In the given circuit, the charge on $4\mu\text{F}$ capacitor will be :



A. $24 \mu\text{C}$

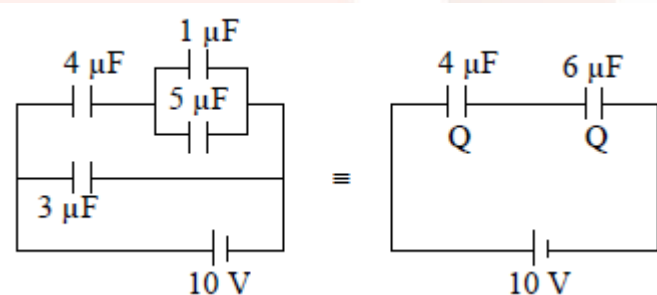
B. $5.4 \mu\text{C}$

C. $9.6 \mu\text{C}$

D. $13.4 \mu\text{C}$

Ans: A

Given circuit is



So, potential difference across upper branch = 10V

In parallel, $C_{\text{eq}} = 5 + 1 = 6 \mu\text{F}$

And in series, $C_{\text{eq}} = 2.4 \mu\text{F}$

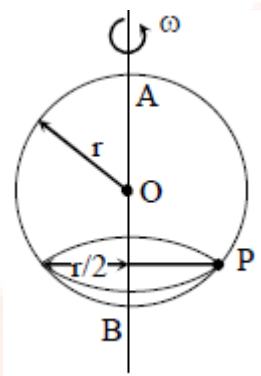
Now $Q = C \times V$

$$= C_{\text{eq}} \times V = 2.4 \mu\text{F} \times 10 = 24 \mu\text{C}$$

Charge of $4\mu\text{F}$ and $6\mu\text{F}$ capacitor would be same

$$Q_{4\mu\text{F}} = 24\mu\text{C}$$

5. A smooth wire of length $2\pi r$ is bent into a circle and kept in a vertical plane. A bead can slide smoothly on the wire. When the circle is rotating with angular speed ω about the vertical diameter AB, as shown in figure, the bead is at rest with respect to the circular ring at position P as shown. Then the value of ω^2 is equal to :



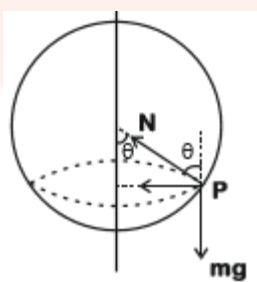
A. $2g / (r\sqrt{3})$

B. $2g / r$

C. $\frac{\sqrt{3}g}{2r}$

D. $(g\sqrt{3}) / r$

Ans: A



$$\theta = \sin^{-1} \frac{1}{2} = 30^\circ$$

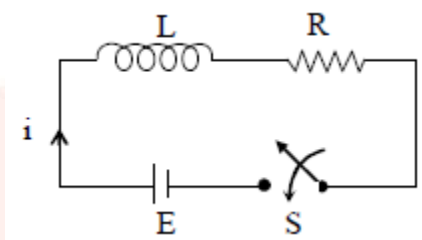
$$N \cos \theta = mg \quad \dots(i)$$

$$N \sin \theta = \frac{m\omega^2 r}{2} \quad \dots(ii)$$

$$\tan \theta = \frac{\omega^2 r}{2g}$$

$$\Rightarrow \omega^2 = \frac{2g \tan \theta}{r} = \frac{2g}{r\sqrt{3}}$$

6. Consider the LR circuit shown in the figure. If the switch S is closed at $t = 0$ then the amount of charge that passes through the battery between $t = 0$ and $t = \frac{L}{R}$ is :



- A. $\frac{EL}{2.7R^2}$
 B. $\frac{EL}{7.3R^2}$
 C. $\frac{7.3EL}{R^2}$
 D. $\frac{2.7EL}{R^2}$

Ans: A

In an L-R circuit, current during charging is given by

$$I = I_{\max} \left(1 - e^{-\frac{Rt}{L}} \right) \quad \text{where} \quad I_{\max} = \frac{E}{R}$$

So, we have

$$\frac{dq}{dt} = \frac{E}{R} \left(1 - e^{-\frac{Rt}{L}} \right)$$

$$\int_0^{\frac{4R}{L}} dq = \frac{E}{R} \int_0^{\frac{4R}{L}} \left(1 - e^{-\frac{Rt}{L}} \right) dt$$

$$Q = \frac{E}{R} \left[L + \frac{L}{R} e^{-\frac{Rt}{L}} \right]_0^{L/R}$$

$$= \frac{E}{R} \left[\frac{L}{R} + \frac{L}{R} e^{-1} - \frac{L}{R} \right]$$

$$Q = \frac{EL}{R^2 e} \Rightarrow Q = \frac{EL}{2.7R^2}$$

7. A tuning fork of frequency 480 Hz is used in an experiment for measuring speed of sound ($\mathcal{V}$) in air by resonance tube method. Resonance lengths of the air column, $l_1 = 30$ cm, $l_2 = 70$ cm, Then, $\mathcal{V}$ is equal to :

A. 384 ms^{-1} B. 332 ms^{-1}
C. 379 ms^{-1} D. 338 ms^{-1}

Ans: A

Given that

$$l_1 = 30 \text{ cm}$$

$$l_2 = 70 \text{ cm}, f = 480$$

As we know that speed of sound $v = f\lambda$

$$v = 2(l_2 - l_1)f$$

$$= 2(70 - 30) \times 10^{-2} \times 480$$

$$= 2 \times 40 \times 10^{-2} \times 480$$

$$= 384 \text{ m/sec}$$

8. A solid sphere, of radius R acquires a terminal velocity $\mathcal{V}_1$ when falling (due to gravity) through a viscous fluid having a coefficient of viscosity η . The sphere is broken into 27 identical solid spheres. If each of these spheres acquires a terminal velocity, $\mathcal{V}_2$, when falling through the same fluid, the ratio $(\mathcal{V}_1 / \mathcal{V}_2)$ equals:

A. $1/9$ B. 9
C. 27 D. $1/27$

Ans: B

Volume of sphere of radius $R = 27 \times$ volume of sphere of radius r

$$\frac{4}{3} \pi R^3 = 27 \times \frac{4}{3} \pi r^3$$

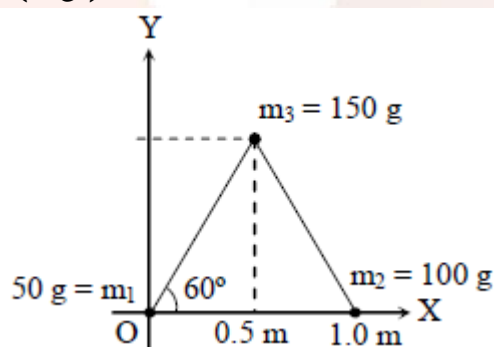
$$\Rightarrow r^3 = \frac{R^3}{3^3} \Rightarrow r = \frac{R}{3}$$

$$V_1 = \frac{(\rho_0 - \rho_{\text{liq}}) \frac{4}{3} \pi R^3 g}{6\pi\eta R}$$

$$V_2 = \frac{(\rho_0 - \rho_{\text{liq}}) \frac{4}{3} \pi \left(\frac{R}{3}\right)^3 g}{6\pi\eta \left(\frac{R}{3}\right)} = \frac{(\rho_0 - \rho_{\text{liq}}) \frac{4}{3} \pi R^3 g \times \frac{1}{27}}{6\pi\eta R \times \frac{1}{3}}$$

$$V_2 = \frac{V_1}{9} \Rightarrow \frac{V_1}{V_2} = 9$$

9. Three particles of masses 50 g, 100 g and 150 g are placed at the vertices of an equilateral triangle of side 1 m (as shown in the figure). The (x, y) coordinates of centre of mass will be :



- A. $\left(\frac{\sqrt{3}}{8} \text{ m}, \frac{7}{12} \text{ m}\right)$ B. $\left(\frac{7}{12} \text{ m}, \frac{\sqrt{3}}{8} \text{ m}\right)$
 C. $\left(\frac{\sqrt{3}}{4} \text{ m}, \frac{5}{12} \text{ m}\right)$ D. $\left(\frac{7}{12} \text{ m}, \frac{\sqrt{3}}{4} \text{ m}\right)$

Ans: D

Given that

$$m_1 = 50g \quad m_2 = 100g \quad m_3 = 150g$$

$$x_1 = 0 \quad x_2 = 1m \quad x_3 = 0.5m$$

$$y_1 = 0 \quad y_2 = 0 \quad y_3 = \frac{\sqrt{3}}{2}$$

$$x_{COM} = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3} = \frac{0 + (100)(1) + (150)(0.5)}{300}$$

$$x_{COM} = \frac{100 + 75}{300} = \frac{175}{300} = \frac{7}{12}m.$$

$$y_{COM} = \frac{m_1y_1 + m_2y_2 + m_3y_3}{m_1 + m_2 + m_3} = \frac{0 + (100)(0) + (150)\frac{\sqrt{3}}{2}}{300}$$

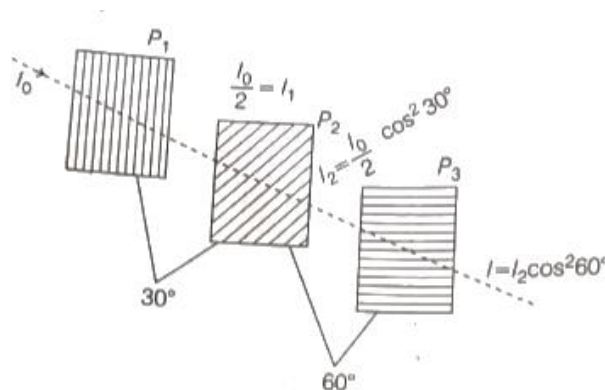
$$y_{COM} = \frac{75\sqrt{3}}{300} = \frac{3\sqrt{3}}{12} = \frac{\sqrt{3}}{4}m$$

$$(x_{COM}, y_{COM}) = \left(\frac{7}{12}, \frac{\sqrt{3}}{4} \right)$$

10. A system of three polarizers P_1, P_2, P_3 is set up such that the pass axis of P_3 is crossed with respect to that of P_1 . The pass axis of P_2 is inclined at 60° to the pass axis of P_3 . When a beam of unpolarized light of intensity I_0 is incident on P_1 , the intensity of light transmitted by the three polarizers is I . The ratio (I_0/I) equals (nearly) :

- A. 10.67 B. 5.33
C. 1.80 D. 16.00

Ans: A



When unpolarised light pass through polaroid P_1 , intensity obtained is

$$I_1 = \frac{I_0}{2}$$

Where, I_0 = intensity of incident light.

Now, this transmitted light is polarised and it pass through polariser P_2 . So, intensity I_2 transmitted is obtained by Malus law.

$$\Rightarrow I_2 = I_1 \cos^2 \theta$$

As angle of pass axis of P_1 and P_3 is 90° and angle of pass axis of P_2 and P_3 is 60° , so angle between pass axis of P_1 and P_2 is $(90^\circ - 60^\circ) = 30^\circ$. So,

$$I_2 = \frac{I_0}{2} \cos^2 30^\circ = \frac{I_0}{2} \times \left(\frac{\sqrt{3}}{2} \right)^2 = \frac{3}{8} I_0$$

When this light pass through third polarise P_3 , intensity I transmitter is again obtained by Malus law.

$$\text{So, } I = I_2 \cos^2 60^\circ = \left(\frac{3}{8} I_0 \right) \cos^2 60^\circ$$

$$= \frac{3}{8} I_0 \times \left(\frac{1}{2} \right)^2 = \frac{3}{32} I_0$$

$$\text{So, ratio } \frac{I_0}{I} = \frac{32}{3} = 10.67$$

11. The ratio of the weights of a body on the Earth's surface to that on the surface of a planet is 9 : 4. The mass of the planet is $\frac{1}{9}$ th of that of the Earth. If 'R' is the radius of the Earth, what is the radius of the planet ? (Take the planets to have the same mass density).

- A. $\frac{R}{2}$ B. $\frac{R}{4}$
C. $\frac{R}{9}$ D. $\frac{R}{3}$

Ans: A

Given that

$$\frac{g_e}{g_p} = \frac{9}{4} \text{ and } M_p = \frac{1}{9} M_e$$

$$\Rightarrow \frac{M_e \times R_p^2}{R_e^2 \times M_p} = \frac{9}{4}$$

$$\Rightarrow 9 \times \left(\frac{R_p}{R_e} \right)^2 = \frac{9}{4} \Rightarrow R_p = \frac{R_e}{2}$$

$$\therefore R_p = \frac{R}{2}$$

12. In an amplitude modulator circuit, the carrier wave is given by, $C(t) = 4 \sin(20000 \pi t)$ while modulating signal is given by, $m(t) = 2 \sin(2000 \pi t)$. The values of modulation index and lower side band frequency are :

- A. 0.5 and 10 kHz B. 0.4 and 10 kHz
C. 0.3 and 9 kHz D. 0.5 and 9 kHz

Ans: D

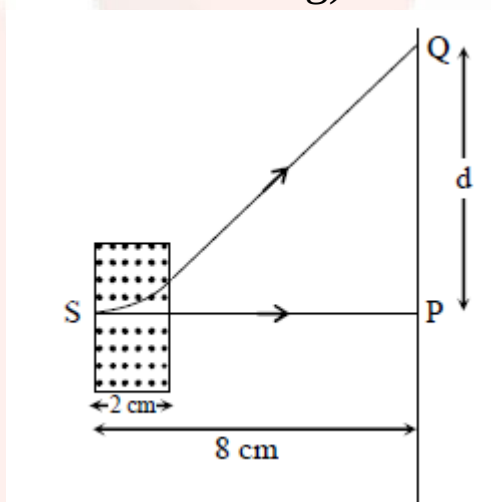
$$C(t) = 4 \sin(20000 \pi t), A_c = 4, f_c = \frac{\omega_c}{2\pi} = \frac{20000\pi}{2\pi} = 10,000$$

$$m(t) = 2 \sin(2000\pi t), A_m = 2, f_m = \frac{\omega_m}{2\pi} = \frac{2000\pi}{2\pi} = 1000$$

Modulation index $\mu = \frac{A_m}{A_c} = \frac{2}{4} = \frac{1}{2} = 0.5$

Lower side band frequency = $f_c - f_m$
 $= 10,000 - 1000 = 9000 = 9 \text{ kHz}$

13. An electron, moving along the X -axis with an initial energy of 100 eV, enters a region of magnetic field $\vec{B} = (1.5 \times 10^{-3} \text{ T}) \hat{k}$ at S (see figure). The field extends between $X = 0$ and $X = 2 \text{ cm}$. The electron is detected at the point Q on a screen placed 8 cm away from the point S. The distance between P and Q (on the screen) is: (electron's charge = $1.6 \times 10^{-19} \text{ C}$, mass of electron = $9.1 \times 10^{-31} \text{ kg}$)



- A. 1.22 cm B. 11.65 cm
 C. 2.25 cm D. 12.87 cm

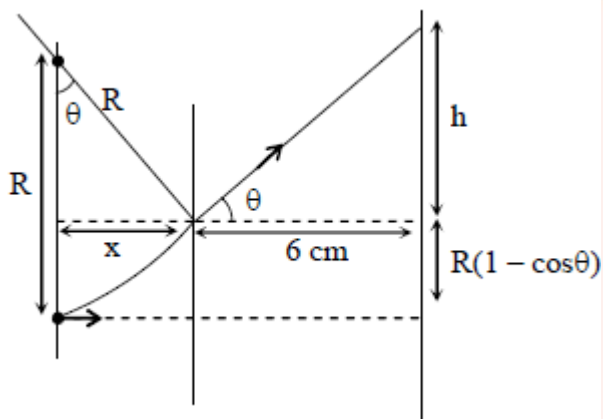
Ans: D

Radius of path, $R = \frac{M_V}{e_B}$

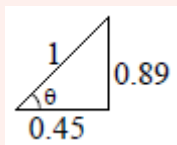
$$R = \frac{\sqrt{2mk}}{qB}$$

$$= \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times 100 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times 1.5 \times 10^{-3}}$$

$$R = 2.24 \text{ cm}$$



$$\sin \theta = \frac{x}{R} = \frac{2}{2.24} = 0.89$$



$$\tan \theta = \frac{0.89}{0.45} \approx 1.97 = \frac{h}{6}$$

$$h = 11.82 \text{ cm}$$

$$\cos \theta = 0.45$$

$$d = h + 2(1 - 0.45) = 11.82 + 2(1 - 0.45) \approx 12.9 \text{ cm}$$

14. A diatomic gas with rigid molecules does 10 J of work when expanded at constant pressure. What would be the heat energy absorbed by the gas, in this process?
- A. 40 J B. 30 J
C. 35 J D. 25 J

Ans: C

Given that

$W = 10 \text{ J}$ at constant pressure

Work done by a gas,

$$10 = nR(T_2 - T_1) = nR\Delta T$$

Heat absorbed by gas at constant pressure

$$\Delta Q = nC_p\Delta T = \frac{7}{2}Rn\Delta T = \frac{7}{2} \times 10 \quad \left(\because C_p = \frac{7}{2}R \text{ for diatomic gas} \right)$$

$$= 35 \text{ J}$$

15. A carnot engine has an efficiency of $1/6$, When the temperature of the sink is reduced by 62° C , its efficiency is doubled. The temperatures of the source and the sink are, respectively,

A. 99° C , 37° C B. 62° C , 124° C
 C. 37° C , 99° C D. 124° C , 62° C

Ans: C

Given that

Efficiency of carnot engine

$$\eta = \frac{1}{6}$$

Efficiency of a carnot engine working between source of temperature T_H and sink of temperature T_L is given by

$$\frac{1}{6} = 1 - \frac{T_L}{T_H} \quad \dots(1)$$

$$\frac{1}{3} = 1 - \frac{(T_L - 62)}{T_H} \quad \dots(2)$$

Solving eq. (1)

$$\Rightarrow \frac{1}{6} = \frac{T_H - T_L}{T_H}$$

$$\Rightarrow T_H = 6T_H - 6T_L$$

$$6T_L = 5T_H$$

$$T_H = \frac{6T_L}{5}$$

Solving eq. (2)

$$\frac{1}{3} = \frac{T_H - (T_L - 62)}{T_H} \Rightarrow T_H = 3T_H - 3T_L + 186$$

$$\Rightarrow 2T_H = 3T_L - 186$$

$$2 \times \frac{6T_L}{5} = 3T_L - 186$$

$$\Rightarrow 12T_L = 15T_L - 930 \Rightarrow 3T_L = 930$$

$$T_L = 310K$$

$$T_L = 310 - 273 = 37^\circ C$$

Source temp. is higher & sink temp. is lower

$$T_H = \frac{6T_L}{5} = \frac{6 \times 310}{5} = 372K = 99^\circ C$$

16. A plane electromagnetic wave having a frequency $\nu = 23.9$ GHz propagates along the positive z – direction in free space. The peak value of the Electric Field is 60 C/m. Which among the following is the acceptable magnetic field component in the electromagnetic wave ?

A. $\vec{B} = 2 \times 10^{-7} \sin(1.5 \times 10^2 x + 0.5 \times 10^{11} t) \hat{j}$

B. $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^2 z + 1.5 \times 10^{11} t) \hat{i}$

C. $\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$

D. $\vec{B} = 60 \sin(0.5 \times 10^2 x + 1.5 \times 10^{11} t) \hat{i}$

Ans: C

Given that

$$\nu = 23.9 \text{ GHz} = 23.9 \times 10^9 \text{ Hz}$$

$$E_0 = 60 \text{ V/m}$$

∴ As we know that

$$\frac{E_0}{B_0} = C \Rightarrow B_0 = \frac{E_0}{C} = \frac{60}{3 \times 10^8} = 2 \times 10^{-7}$$

$$\omega = 2\pi\nu = 2 \times 3.142 \times 23.9 \times 10^9 = 1.5 \times 10^{11} \text{ s}^{-1}$$

$$K = \frac{\omega}{C} = \frac{1.5 \times 10^{11}}{3 \times 10^8} = 0.5 \times 10^3$$

$$\vec{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t) \hat{i}$$

17. A uniform cylindrical rod of length L and radius r , is made from a material whose Young's modulus of Elasticity equals Y . When this rod is heated by temperature T and simultaneously subjected to a net longitudinal compressional force F , its length remains unchanged. The coefficient of volume expansion, of the material of the rod, is nearly) equal to:

- A. $6F / (\pi r^2 Y T)$ B. $9F / (\pi r^2 Y T)$
C. $3F / (\pi r^2 Y T)$ D. $F / (3 \pi r^2 Y T)$

Ans: C

Now, compression strain is obtained by using formula for young is modulus.

$$Y = \frac{\frac{F}{A}}{\frac{\Delta \ell}{\ell}} = \frac{\frac{F}{\pi r^2}}{\frac{\Delta \ell}{\ell}} \Rightarrow Y = \frac{F}{\pi r^2} \times \frac{\ell}{\Delta \ell}$$

$$\Delta \ell = \frac{F \ell}{\pi r^2 Y}$$

Change in length due to temperature change is obtained by using formula for expansion in red

$$\Delta l = l\alpha\Delta T$$

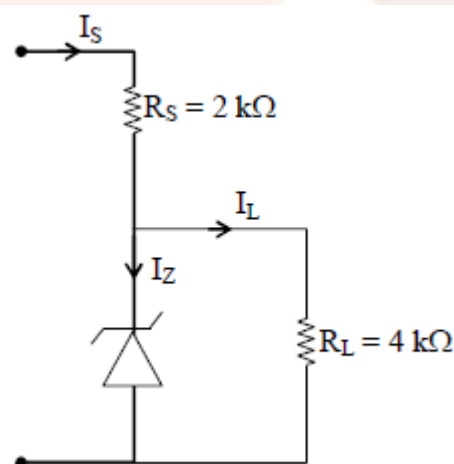
$$l\alpha T = \frac{F l}{\pi r^2 Y}$$

$$\Rightarrow \alpha = \frac{F}{\pi r^2 Y T}$$

The coefficient of volume expansion, $Y = 3\alpha$

$$Y = \frac{3F}{\pi r^2 Y T}$$

18. Figure shows a DC voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is the maximum Zener current?



- A. 1.5 mA B. 7.5 mA
C. 2.5 mA D. 3.5 mA

Ans: D

I_Z is maximum when input voltage is 16 V.

$$I_S = \frac{10}{2 \times 10^3} = 5 \text{ mA}$$

$$I_L = \frac{6}{4 \times 10^3} = 1.5 \text{ mA}$$

$$I_{Z(\text{max})} = I_S - I_L = 3.5 \text{ mA}$$

19. One kg of water, at 20°C , is heated in an electric kettle whose heating element has a mean (temperature averaged) resistance of $20\ \Omega$. The rms voltage in the mains is 200 V . Ignoring heat loss from the kettle, time taken for water to evaporate fully, is close to:

[Specific heat of water = $4200\text{ J/kg }^{\circ}\text{C}$), Latent heat of water = 2260 kJ/kg]

- A. 10 minutes B. 16 minutes
C. 3 minutes D. 22 minutes

Ans: D

Given that

$$R = 20\ \Omega \text{ and } P = \frac{V^2}{R} = \frac{200 \times 200}{20} = 2000 \text{ watt}$$

$$V = 200\text{ V}$$

$$1\text{ kg water } 20^{\circ}\text{C} \rightarrow 1\text{ kg water } 100^{\circ}\text{C}$$

$$\text{Heat required } Q_1 = ms\Delta T = (1)(4200)(80) = 336000$$

$$1\text{ kg water } 100^{\circ}\text{C} \rightarrow 1\text{ kg vapour}$$

$$\text{Heat required } Q_2 = mL = (1) 2260 \times 1000 = 2260000$$

Heat required by water for getting not and then evaporated is

$$Q = Q_1 + Q_2$$

$$Pt = Q_1 + Q_2$$

$$\Rightarrow 2000 \times t = 2260000 + 336000$$

$$t = 1298 \text{ sec.}$$

$$t = 21.63 \text{ mint.} \approx 22 \text{ minute}$$

20. Half lives of two radioactive nuclei A and B are 10 minutes and 20 minutes, respectively. If, initially a sample has equal number of nuclei, then after 60 minutes, the ratio of decayed numbers of nuclei A and B will be:

- A. 3: 8 B. 1: 8
C. 9: 8 D. 8: 1

Ans: C

Number of active nuclei remained after 60 min can also be calculated as

$$N' = \frac{N}{2^{T/t_1/t_2}} \text{ where, } T = 60 \text{ min}$$

So, for nuclei A

$$A_1 = \frac{A_0}{2^{60/10}} = \frac{A_0}{2^6} = \frac{A_0}{64}$$

for nuclei B

$$A_2 = \frac{A_0}{2^{60/20}} = \frac{A_0}{2^3} = \frac{A_0}{8}$$

$$\text{No. of decayed nuclei} = A'_1 = A_0 - \frac{A_0}{64} = \frac{63}{64}$$

$$A'_2 = A_0 - \frac{A_0}{8} = \frac{7}{8}$$

$$\text{Required Ratio} = \frac{63}{64} \times \frac{8}{7} = \frac{9}{8}$$

21. Two particles are projected from the same point with the same speed u such that they have the same range R , but different maximum heights, h_1 and h_2 . Which of the following is correct?

- A. $R^2 = 4 h_1 h_2$ B. $R^2 = 16 h_1 h_2$
C. $R^2 = h_1 h_2$ D. $R^2 = 2 h_1 h_2$

Ans: B

At complementary angles, ranges are equal.

$$h_1 = \frac{u^2 \sin^2 \theta}{2g}, h_2 = \frac{u^2 \cos^2 \theta}{2g}$$

$$R = \frac{2u^2 \sin \theta \cos \theta}{g}$$

$$h_1 h_2 = \frac{u^4 \sin^2 \theta \cos^2 \theta}{4g^2}$$

$$R^2 = \frac{4u^4 \sin^2 \theta \cos^2 \theta}{g^2}$$

$$R^2 = 16h_1 h_2$$

22. A particle is moving with speed $v = b\sqrt{x}$ along positive x – axis. Calculate the speed of the particle at time $t = \tau$ (assume that the particle is at origin at $t = 0$).

A. $b^2\tau$

B. $\frac{b^2\tau}{\sqrt{2}}$

C. $\frac{b^2\tau}{2}$

D. $\frac{b^2\tau}{4}$

Ans: B

Given, speed

$$v = b\sqrt{x} \quad \dots(i)$$

Now, differentiating it with respect to time, we get

$$\frac{dx}{dt} = bx^{1/2} \Rightarrow \int_0^x x^{-1/2} dx = \int_0^\tau b dt$$

$$\Rightarrow 2\sqrt{x} = b\tau \Rightarrow \sqrt{x} = \frac{b\tau}{2}$$

from equation (i) we get

$$v = b \frac{b\tau}{2} = \frac{b^2\tau}{2}$$

$$v = \frac{b^2\tau}{2}$$

23. The number density of molecules of a gas depends on their distance r from the origin as, $n(r) = n_0 e^{-\alpha r^4}$. Then the total number of molecules is proportional to:

- A. $\sqrt{n_0} \alpha^{1/2}$ B. $n_0 \alpha^{1/4}$
C. $n_0 \alpha^{-3}$ D. $n_0 \alpha^{-3/4}$

Ans:

Given that

$$n = n_0 e^{-\alpha r^4}$$

$$\Rightarrow N = 4\pi n_0 \cdot \int_0^{\infty} r^2 e^{-\alpha r^4} dr$$

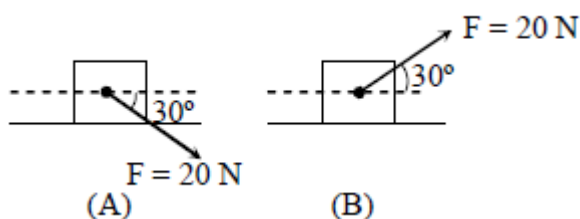
$$\text{Put } \sqrt{\alpha} r^2 = t$$

$$N = \frac{4\pi n_0}{2\sqrt{\alpha}} \int_0^{\infty} \frac{t^{\frac{1}{2}} e^{-t^2}}{\alpha^{\frac{1}{4}}} dt$$

$$= \frac{4\pi n_0}{2\alpha^{\frac{3}{4}}} \int_0^{\infty} t^{\frac{1}{2}} e^{-t^2} dt$$

$$N \propto n_0 \alpha^{-\frac{3}{4}}$$

24. A block of mass 5 kg is (i) pushed in case (A) and (ii) pulled in case (B), by a force $F = 20$ N, making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $\mu = 0.2$ the difference between the accelerations of the block, in case (B) and case (A) will be : ($g = 10 \text{ ms}^{-2}$)

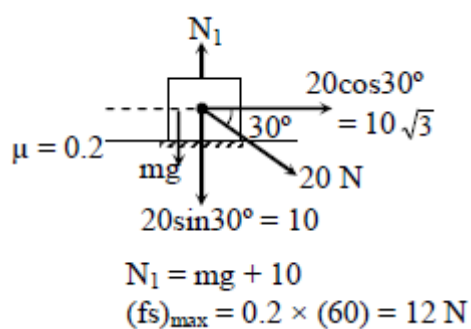


- A. 0.4 ms^{-2} B. 0.8 ms^{-2}
C. 3.2 ms^{-2} D. 0 ms^{-2}

Ans: B

Case-A

Free body diagram of block is

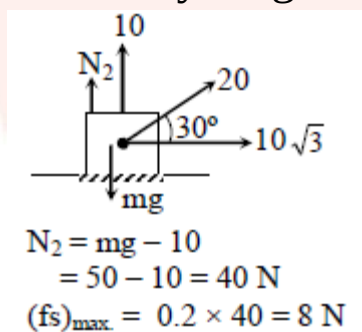


$$a_A = \frac{10\sqrt{3} - 12}{5} = \frac{17.32 - 12}{5}$$

$$a_A = \frac{5.32}{5}$$

Case – B

Free body diagram of block is



$$a_B = \frac{10\sqrt{3} - 8}{5} = \frac{17.32 - 8}{5}$$

$$a_B = \frac{9.32}{5}$$

Difference between acceleration $a_B - a_A = \frac{1}{5}(9.32 - 5.32) = \frac{4}{5}$

$$\Delta a = 0.8 \text{ m / s}^2$$

25. Let a total charge $2Q$ be distributed in a sphere of radius R , with the charge density given by $\rho(r) = kr$, where r is the distance from the centre. Two charges A and B , of $-Q$ each, are placed on diametrically opposite points, at equal distance, a from the centre. If A and B do not experience any force then.

- A. $a = 2^{-1/4}R$ B. $a = \frac{3R}{2^{1/4}}$
C. $a = 8^{-1/4}R$ D. $a = R / \sqrt{3}$

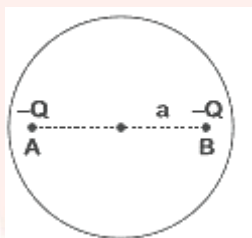
Ans: C

According to Gauss's law

$$\Phi = \frac{q_{in}}{\epsilon_0} \text{ where } \Phi = \text{Electric Flux}$$

$$E \cdot A = \frac{q_{in}}{\epsilon_0} \text{ where } E = \text{Electric Field and } A = \text{Surface Area}$$

$$E \times 4\pi a^2 = \frac{q_{in}}{\epsilon_0}$$



$$q_{in} = \int_0^a (kr) \times 4\pi r^2 dr$$

$$= 4\pi k \cdot \left(\frac{a^4}{4} \right)$$

$$\therefore E \times 4\pi a^2 = \frac{4\pi k a^4}{4\epsilon_0} \Rightarrow E = \frac{k a^2}{4\epsilon_0}$$

$$\begin{aligned}\therefore \frac{ka^2}{4\epsilon_0} \times Q &= \frac{Q^2}{4\pi\epsilon_0 \times 4a^2} \\ \frac{2Q}{\pi R^4} \cdot \frac{a^2 Q}{4\epsilon_0} &= \frac{Q^2}{4\pi\epsilon_0 \times 4a^2} \Rightarrow 8a^4 = R^4 \\ \Rightarrow 8^{1/4} \cdot a &= 4 \\ \Rightarrow a &= \frac{R}{8^{1/4}}\end{aligned}$$

26. A spring whose unstretched length is l has a force constant k . The spring is cut into two pieces of unstretched lengths l_1 and l_2 where, $l_1 = nl_2$ and n is an integer. The ratio k_1/k_2 of the corresponding force constants, k_1 and k_2 will be :

- A. $\frac{1}{n^2}$ B. n
C. n^2 D. $\frac{1}{n}$

Ans: D

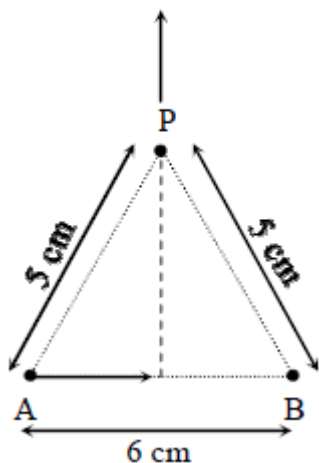
Given that

$$L_1 = nl_2$$

$$\therefore K \propto \frac{1}{l}$$

$$\therefore \frac{K_1}{K_2} = \frac{l_2}{l_1} = \frac{1}{n}$$

27. Find the magnetic field at point P due to a straight line segment AB of length 6 cm carrying a current of 5 A. (see figure) ($\mu_0 = 4\pi \times 10^{-7} \text{ N - A}^{-2}$)



- A. $1.5 \times 10^{-5} \text{ T}$ B. $2.5 \times 10^{-5} \text{ T}$
C. $2.0 \times 10^{-5} \text{ T}$ D. $3.0 \times 10^{-5} \text{ T}$

Ans: A

By Biot-Savart's law, magnetic field due to a wire segment at point p is given by

$$B = \frac{\mu_0 I}{4\pi d} (\sin \theta_1 + \sin \theta_2)$$

From given data

$$I = 5 \text{ A}$$

$$\mu = 4\pi \times 10^{-7} \text{ N A}^{-2}$$

$$d = \sqrt{5^2 - 3^2} = \sqrt{16} = 4 \text{ cm}$$

$$\sin \theta_1 = \sin \theta_2 = \frac{3}{5}$$

$$= \frac{5}{4 \times 10^{-2}} \left(\frac{3}{5} + \frac{3}{5} \right) \times 10^{-7} \quad 4 \text{ cm} = 4 \times 10^{-2} \text{ m}$$

$$= \frac{5}{4} \times 2 \frac{3 \times 10^{-7}}{5 \times 10^{-2}}$$

$$B = 1.5 \times 10^{-5} \text{ T}$$

28. Consider an electron in a hydrogen atom, revolving in its second excited state (having radius $4.65 \text{ \AA}$). The de-Broglie wavelength of this electron is :

- A. $3.5 \text{ \AA}$ B. $9.7 \text{ \AA}$
C. $6.6 \text{ \AA}$ D. $12.9 \text{ \AA}$

Ans: B

As we know that

$n = 3$ (second excited state)

As we know that de-Broglie wavelength associated with it, is given by

$$2\pi r_n = n\lambda_{dB}$$

Given, $n = 3$, $r_3 = 4.65 \text{ \AA}$

$$\Rightarrow \lambda_{dB} = \frac{2\pi r_3}{n} = \frac{2 \times 3.14 \times 4.65}{3} \approx 9.7 \text{ \AA}$$

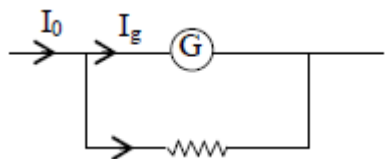
29. A moving coil galvanometer, having a resistance G , produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then,

- A. $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$
B. $R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g} \right)^2$
C. $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{I_0 - I_g}$
D. $R_A R_V = G^2 \left(\frac{I_0 - I_g}{I_g} \right)$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$

Ans: A

To use galvanometer as an ammeter, a low resistance in parallel is used

In ammeter, if I_g = full scale deflection



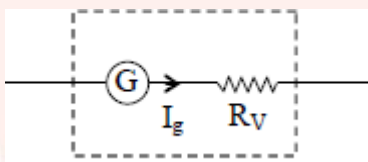
$$I_0 - I_g \cdot S = R_A$$

$$I_g G = (I_0 - I_g) R_A$$

$$R_A = \frac{I_g G}{(I_0 - I_g)} \quad \dots(1)$$

Here, G = resistance of galvanometer coil

When a galvanometer is used as a voltmeter a high resistance (R_V) is series is used.



$$V = I_g(G + R_V)$$

$$GI_0 = I_g(G + R_V)$$

$$R_V = \frac{GI_0}{I_g} - G$$

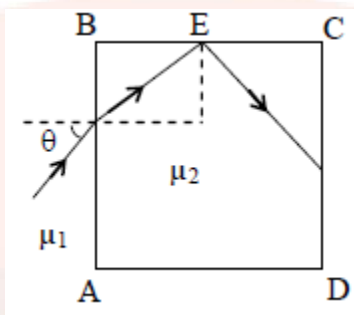
$$R_V = \frac{G(I_0 - I_g)}{I_g} \quad \dots(2)$$

So from (1) & (2)

$$R_A R_V = G^2$$

$$\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$$

30. A transparent cube of side d , made of a material of refractive index μ_2 , is immersed in a liquid of refractive index μ_1 ($\mu_1 < \mu_2$). A ray is incident on the face AB at an angle θ (shown in the figure). Total internal reflection takes place at point E on the face BC.



Then θ must satisfy :

A. $\theta < \sin^{-1} \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$

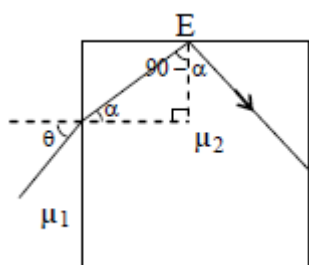
B. $\theta < \sin^{-1} \frac{\mu_1}{\mu_2}$

C. $\theta > \sin^{-1} \frac{\mu_1}{\mu_2}$

D. $\theta > \sin^{-1} \sqrt{\frac{\mu_2^2}{\mu_1^2} - 1}$

Ans: A

In given arrangement



By snell's law

$$\mu_1 \sin \theta = \mu_2 \sin \alpha \dots (1)$$

for total internal reflection, angle of incidence at medium interface must be greater than critical angle.

$$\mu_2 \sin(90 - \alpha) > \mu_1$$

From eq. (1)

$$\sin \alpha = \frac{\mu_1}{\mu_2} \sin \theta$$

$$\cos \alpha = \sqrt{\frac{\mu_2^2 - \mu_1^2 \sin^2 \theta}{\mu_2^2}}$$

$$\cos \alpha = \sqrt{\mu_2^2 - \frac{\mu_1^2 \sin^2 \theta}{\mu_2^2}}$$

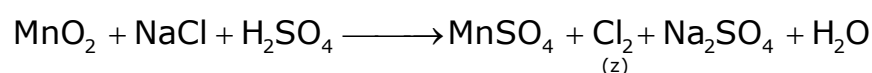
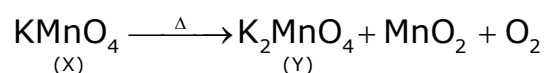
Chemistry

31. Thermal decomposition of a Mn compound (X) at 513 K results in compound Y, MnO_2 reacts with NaCl and concentrated H_2SO_4 to give a pungent gas Z, X, Y, and Z, respectively, are :

- A. K_2MnO_4 , KMnO_4 and Cl_2
- B. KMnO_4 , K_2MnO_4 and Cl_2
- C. K_3MnO_4 , K_2MnO_4 and Cl_2
- D. K_2MnO_4 , KMnO_4 and SO_2

Ans: B

Thermal decomposition of Mn compound (X), i.e., KMnO_4 at 513 K results in compound Y (i.e., K_2MnO_4), MnO_2 and a gaseous product. MnO_2 reacts with NaCl and concentrated H_2SO_4 to give a pungent gas Z (i.e., Cl_2). The reactions involved are as follows:



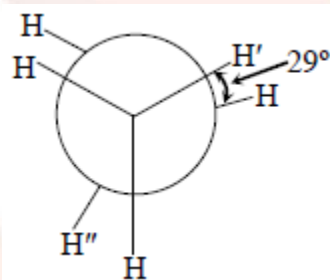
32. Which of the given statements is INCORRECT about glycogen?

- A. It is present in some yeast and fungi.
- B. Only α –linkages are present in the molecule.
- C. It is a straight chain polymer similar to amylose.
- D. It is present in animal cells.

Ans: C

Structure of glycogen is similar to amylopectin glycogen.

33. In the following skew conformation of ethane, $H' - C - C - H''$ dihedral angle is:



- A. 58°
- B. 120°
- C. 151°
- D. 149°

Ans: D

A dihedral angle is the angle between two $C - H$ bonds projected on a plane orthogonal to the $C - C$ bond.

$$H' - C - C - H'' = 120^\circ + 29^\circ = 149^\circ$$

34. NO_2 required for a reaction is produced by the decomposition of N_2O_5 in CCl_4 as per the equation



The initial concentration of N_2O_5 is 3.00 mol L^{-1} and it is 2.75 mol L^{-1} after 30 minutes. The rate of formation of NO_2 is :

A. $1.667 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$

B. $8.333 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$

C. $4.167 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$

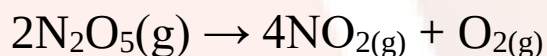
D. $2.083 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$

Ans: A

Given, $[\text{N}_2\text{O}_5]_{\text{initial}} = 3.00 \text{ mol L}^{-1}$

After 30 min, $[\text{N}_2\text{O}_5] = 2.75 \text{ mol L}^{-1}$

Given Reaction is



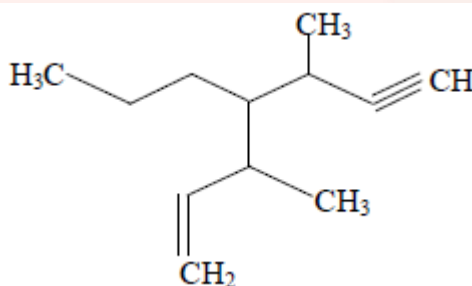
$$\frac{1}{2} \frac{d[\text{N}_2\text{O}_5]}{dt} = -1 \frac{1}{4} \frac{d[\text{NO}_2]}{dt}$$

$$\Rightarrow \frac{d[\text{NO}_2]}{dt} = -2 \frac{d[\text{N}_2\text{O}_5]}{dt}$$

$$\Rightarrow \frac{d[\text{NO}_2]}{dt} = -2 \frac{(3 - 2.75)}{30}$$

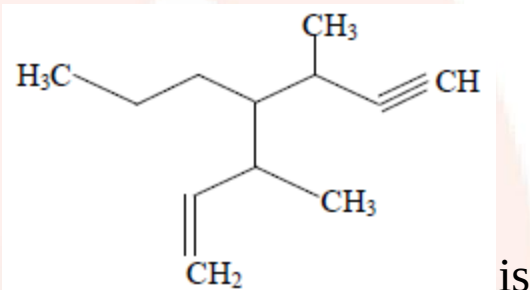
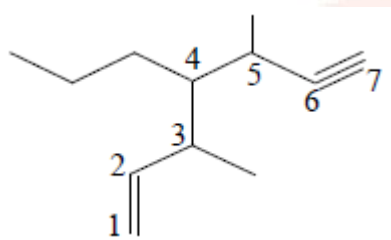
$$\frac{d[\text{NO}_2]}{dt} = \frac{2 \times 0.25}{30} = 1.667 \times 10^{-2} \text{ mol L}^{-1} \text{ min}^{-1}$$

35. The IUPAC name for the following compound is:



- A. 3 – methyl – 4 – (3-methylprop-1-enyl)-1-heptyne
- B. 3,5 – dimethyl – 4 propylhep-1-en-6-yne
- C. 3 – methyl – 4- (1-methylprop-2-ynyl)-1-heptene
- D. 3,5-dimethyl -4-propylhept-6-en-1-1yne

Ans: B



IUPCAC name of the structure

is

– 3,5-dimethyl-4-propylhept-1-en-6-yne

36. An 'Assertion' and a 'Reason' are given below. Choose the correct answer from the following options:

Assertion (A): Vinyl halides do not undergo nucleophilic substitution easily.

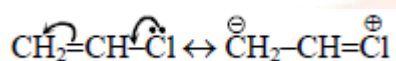
Reason R: Even though the intermediate carbocation is stabilized by loosely held π -electrons, the cleavage is difficult because of strong bonding.

- A. Both (A) and (R) are wrong statements.
- B. Both (A) and (R) are correct statements and (R) is the correct explanation of (A).
- C. (A) is a correct statement but (R) is a wrong statement.

D. Both (A) and (R) are correct statements but (R) is not the correct explanation of (A).

Ans: C

Vinyl halide undergoes nucleophilic substitution reaction. This is because it forms highly unstable carbocation.

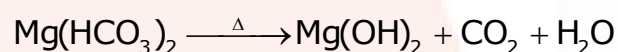


37. The temporary hardness of a water sample is due to compound X. Boiling this sample converts X to compound Y. X and Y, respectively, are :

- A. $\text{Mg}(\text{HCO}_3)_2$ and $\text{Mg}(\text{OH})_2$
- B. $\text{Ca}(\text{HCO}_3)_2$ and $\text{Ca}(\text{OH})_2$
- C. $\text{Mg}(\text{HCO}_3)_2$ and MgCO_3
- D. $\text{Ca}(\text{HCO}_3)_2$ and CaO

Ans: A

Temporary hardness is caused by bicarbonates of calcium and magnesium.

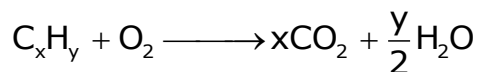


38. 25 g of an unknown hydrocarbon upon burning produces 88 g of CO_2 and 9 g of H_2O . This unknown hydrocarbon contains:

- A. 24 g of carbon and 1 g of hydrogen
- B. 18 g of carbon and 7 g of hydrogen
- C. 20 g of carbon and 5 g of hydrogen
- D. 22 g of carbon and 3 g of hydrogen

Ans: A

Let hydrocarbon is C_xH_y

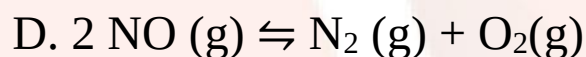
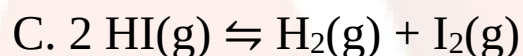
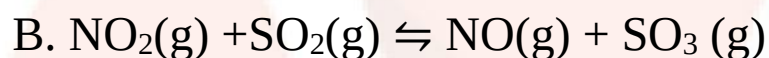
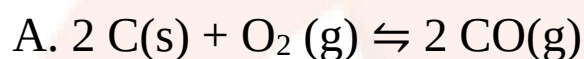


$$\text{wt of carbon} = \frac{88}{44} \times 12 = 24 \text{ g}$$

$$\text{wt of hydrogen} = \frac{9}{18} \times 1 \text{ g}$$

The unknown hydrocarbon contains 24g of carbon and 1 g of hydrogen.

39. In which one of the following equilibria, $K_p \neq K_c$?



Ans: A

The relationship between K_p and K_c

$$K_p = K_c(RT)^{\Delta ng}$$

$$\text{where, } \Delta ng = n_{\text{products}} - n_{\text{reactants}}$$

$$\Delta ng = 0 \text{ so, } K_p = K_c$$

$$\Delta ng \neq 0 \text{ } K_p \neq K_c$$

40. In comparison to boron, beryllium has :

A. Greater nuclear charge and lesser first ionisation enthalpy.

B. Lesser nuclear charge and greater first ionisation enthalpy

C. Greater nuclear charge and greater first ionisation enthalpy

D. Lesser nuclear charge and lesser first ionisation enthalpy.

Ans: B

In comparison to boron, beryllium has lesser nuclear charge and greater first ionisation enthalpy.

41. The compound used in the treatment of lead poisoning is :

- A. EDTA
- B. D – penicillamine
- C. Cis – platin
- D. Desferrioxime B

Ans: A

The compound used in the treatment of lead poisoning is EDTA.

42. A solution is prepared by dissolving 0.6 g of urea (molar mass = 60 g mol^{-1}) and 1.8 g of glucose (molar mass = 180 g mol^{-1}) in 100 mL of water at 27°C . The osmotic pressure of the solution is: ($R = 0.08206 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

- A. 2.46 atm
- B. 8.2 atm
- C. 1.64 atm
- D. 4.92 atm

Ans: D

Given, mass of urea = 0.6 g

Molar mass of urea = 60 g mol^{-1}

Mass of glucose = 1.8g

Molar mass of glucose = 180 g mol^{-1}

Osmotic pressure

$$\pi = i_1 C_1 RT + i_2 C_2 RT$$

$$= (i_1 C_1 + i_2 C_2) RT$$

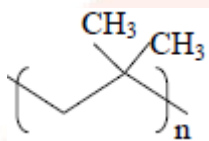
$$= \left[\left(\frac{0.6 \times 1000}{60 \times 1000} \right) + \left(\frac{1.8 \times 1000}{180 \times 1000} \right) \right] RT$$

$$= (0.1 + 0.1) RT$$

$$= 0.2 RT$$

$$= 0.2 \times 0.082 \times 300 = 4.92$$

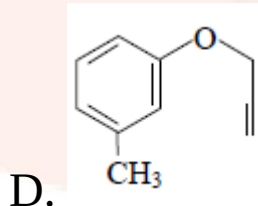
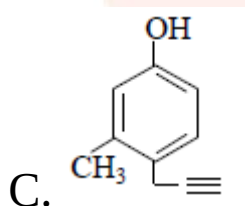
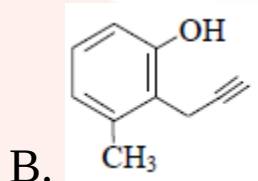
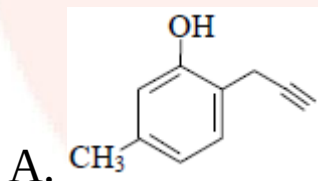
43. The correct name of the following polymer is:



- A. Polyisobutylene B. Polytert-butylene
C. Polyisobutane D. Polyisoprene

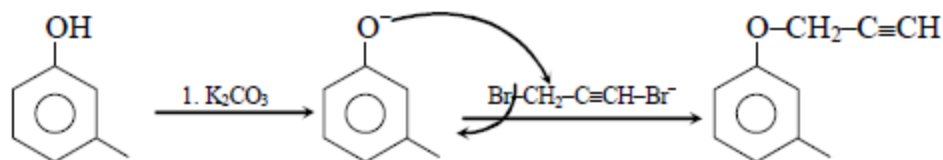
Ans: C

44. What will be the major product when m-cresol is reacted with propargyl bromide ($\text{HC} \equiv \text{C} - \text{CH}_2 \text{Br}$) in presence of K_2CO_3 in acetone ?



Ans: D

The major product when m-cresol reacts with propargyl bromide in presence of K_2CO_3 in acetone is given by

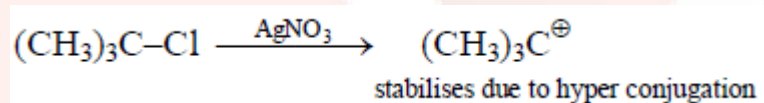


45. Which one of the following is likely to give a precipitate with AgNO_3 solution?

- A. CCl_4 B. $(\text{CH}_3)_3\text{CCl}$
C. $\text{CH}_2 = \text{CH} - \text{Cl}$ D. CHCl_3

Ans: B

$(\text{CH}_3)_3\text{CCl}$ gives a precipitate with AgNO_3 solution because it forms stable carbocation.

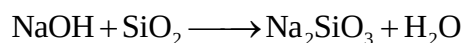
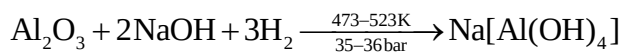


46. The correct statement is:

- A. Leaching of bauxite using concentrated NaOH solution gives sodium aluminate and sodium silicate.
B. Pig iron is obtained from cast iron C. the Hall-Heroult process is used for the production of aluminium and iron.
D. The blistered appearance of copper during the metallurgical process is due to the evolution of CO_2 .

Ans: A

During metallurgy of aluminium, when bauxite (powdered ore) is treated with NaOH (conc.), sodium aluminate ($\text{Na}[\text{Al}(\text{OH})_4]$) is formed and impurity SiO_2 dissolves by forming sodium silicate (Na_2SiO_3)



47. The ratio of number of atoms present in a simple cubic, body centered cubic and face centered cubic structure are, respectively :

- A. 4: 2 : 1
- B. 1: 2: 4
- C. 8: 1: 6
- D. 4: 2: 3

Ans: B

$$\text{Simple cubic unit cell} = \frac{1}{8} \times 8 = 1$$

$$\text{B.C.C. unit cell} = \frac{1}{8} \times 8 + 1 \times 1 = 2$$

$$\text{FCC unit cell} = \frac{1}{8} \times 8 + \frac{1}{2} \times 6 = 4$$

$$\begin{array}{ccc} Z_{\text{SC}} & : & Z_{\text{BCC}} & : & Z_{\text{FCC}} \\ 1 & : & 2 & : & 4 \end{array}$$

48. Among the following, the energy of 2s orbital is lowest in :

- A. H B. Li
- C. Na D. K

Ans: D

Energy is given by

$$E_H = \frac{-13.6Z^2}{n^2}$$

As the value of Z (atomic number) increases, energy of orbitals decreases

Therefore, order of energy of $2s$ orbital is $H > Li > Na > K$

49. The primary pollutant that leads to photochemical smog is:

- | | |
|--------------------|--------------------|
| A. Ozone | B. Acrolein |
| C. Nitrogen oxides | D. Sulphur dioxide |

Ans: D

The primary pollutant that leads to Photochemical smog is Nitrogen oxides particularly they are nitric oxide and nitrogen dioxide.

50. The molar solubility of $Cd(OH)_2$ is $1.84 \times 10^{-5} M$ in water. The expected solubility of $Cd(OH)_2$ in a buffer solution of $pH = 12$ is :

- | | |
|-----------------------------|---|
| A. $2.49 \times 10^{-10} M$ | B. $6.23 \times 10^{-11} M$ |
| C. $1.84 \times 10^{-9} M$ | D. $\frac{2.49}{1.84} \times 10^{-9} M$ |

Ans: A

Given that

The molar solubility of $Cd(OH)_2 = 1.84 \times 10^{-5} M$

Given $pH = 12$

As know that

$$pH + pOH = 14$$

$$12 + pOH = 14$$

$$pOH = 2$$

$[\text{OH}^-] = 10^{-2}$ for Buffer solution

For Reaction $\text{Cd}(\text{OH})_2 \rightarrow \text{Cd}^{2+} + 2\text{OH}^-$
S 2S

$$K_{\text{SP}} = [\text{Cd}^{2+}] [\text{OH}^-]^2$$

$$[\text{Cd}^{2+}] = \frac{K_{\text{SP}}}{[\text{OH}^-]^2} = \text{solubility in buffer solution} \dots (1)$$

while

$$K_{\text{SP}} = 4S^3 \text{ for } \text{Cd}(\text{OH})_2$$

$$\Rightarrow K_{\text{SP}} = 4 \times (1.84 \times 10^{-5})^3 \dots (2)$$

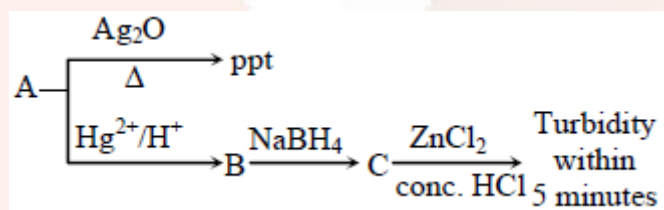
So solubility in Buffer solution is:

$$[\text{Cd}^{2+}] = \frac{K_{\text{SP}}}{[\text{OH}^-]^2} = \frac{4 \times (1.84 \times 10^{-5})^3}{(10^{-2})^2} = 24.9 \times 10^{-11}$$

$$[\text{Cd}^{2+}] = 24.9 \times 10^{-11}$$

$$\text{Solubility} = 2.49 \times 10^{-10}$$

51. Consider the following reactions:



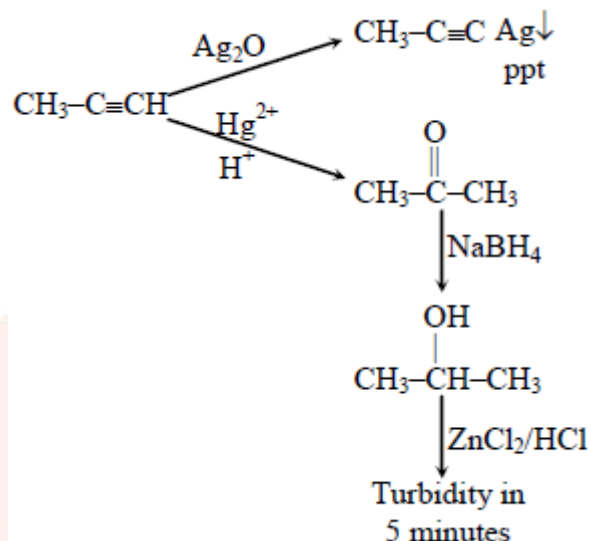
'A' is :

A. $\text{CH}_3 - \text{C} \equiv \text{C} - \text{CH}_3$ B. $\text{CH} \equiv \text{CH}$

C. $\text{CH}_2 = \text{CH}_2$ D. $\text{CH}_3 - \text{C} \equiv \text{CH}$

Ans: D

The reaction undergoes as following:

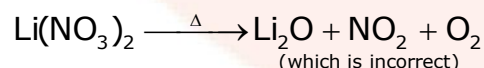


52. The INCORRECT statement is:

- A. LiNO_3 decomposes on heating to give LiNO_2 and O_2 .
- B. Lithium is least reactive with water among the alkali metals.
- C. Lithium is the strongest reducing agent among the alkali metals.
- D. LiCl crystallises from aqueous solution as $\text{LiCl}\cdot 2\text{H}_2\text{O}$.

Ans: A

Statement (A) is incorrect. LiNO_3 on heating gives a mixture of Li_2O , NO_2 and O_2 .



53. The coordination numbers of Co and Al in $[\text{Co}(\text{Cl})(\text{en})_2]\text{Cl}$ and $\text{K}_3[\text{Al}(\text{C}_2\text{O}_4)_3]$, respectively, are :

(en = ethane-1, 2-diamine)

A. 5 and 3 B. 6 and 6

C. 3 and 3 D. 5 and 6

Ans: D



en $\rightarrow$ bidentate
Cl – monodentate

So coordination numbers of Co is 5.



$\text{C}_2\text{O}_4^{2-} \rightarrow$ is bidentate so coordination numbers of Al is 6.

54. The INCORRECT match in the following is :

A. $\Delta G^0 < 0$, $K > 1$

B. $\Delta G^0 < 0$, $K < 1$

C. $\Delta G^0 > 0$, $K < 1$

D. $\Delta G^0 = 0$, $K = 1$

Ans: B

As we know that for Ideal gas

$$\Delta G = \Delta G^0 + RT \ln \theta_1$$

at equilibrium $\Delta G = 0$

$$\Delta G^0 = -2.303 RT \log K$$

if $\Delta G^0 < 0$

$$\Rightarrow -2.303 RT \log K < 0$$

$$\Rightarrow \log K > 0$$

$$\Rightarrow K > 1$$

55. The C – C bond length is maximum in:

- A. C₆₀ B. Graphite
C. Diamond D. C₇₀

Ans: C

The C – C bond length is maximum in diamond.

56. The pair that has similar atomic radii is :

- A. Mo and W B. Mn and Re
C. Ti and Hf D. Sc and Ni

Ans: A

The pair that has similar atomic radii is Mo and W because of lanthanoid contraction.

57. Among the following, the INCORRECT statement about colloids is:

- A. They are larger than small molecules and have high molar mass.
B. They can scatter light.
C. The osmotic pressure of a colloidal solution is of higher order than the true solution at the same concentration.
D. The range of diameters of colloidal particles is between 1 and 1000 nm.

Ans: C

Osmotic pressure of colloidal solution is lower than true solution of same concentration because of association of solute molecule till they acquire colloidal dimensions.

58. The decreasing order of electrical conductivity of the following aqueous solution is:

0.1 M Formic acid (A),

0.2 0.1 M Acetic acid (B),

0.3 0.1 M Benzoic acid (C).

A. $A > B > C$

B. $C > B > A$

C. $A > C > B$

D. $C > A > B$

Ans: C

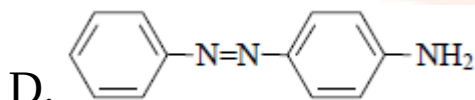
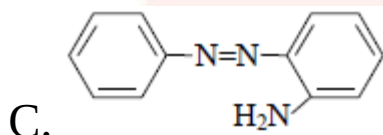
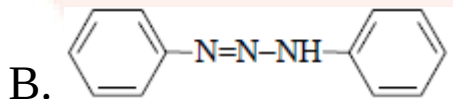
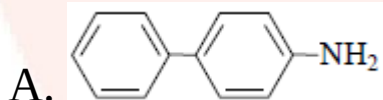
Electrical conductivity is the measurement of electric current, which can be a content or the ability to an electric current.

So, if a aqueous solution is more acidic then it able to conduct electricity.

Order of acidity: $A > C > B$

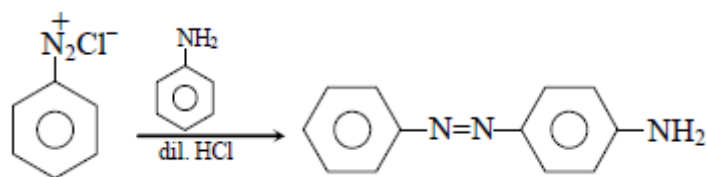
Order of electrical conductivity: $A > C > B$

59. Benzene diazonium chloride on reaction with aniline in the presence of dilute hydrochloric acid gives:

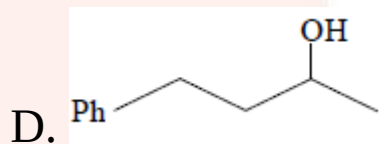
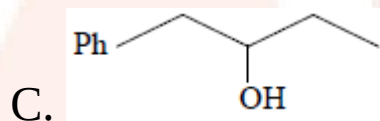
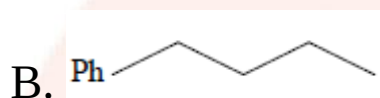
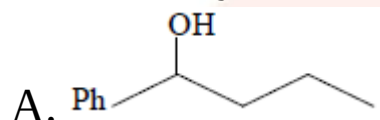


Ans: D

Benzene diazonium chloride on reaction with aniline in the presence of dilute hydrochloric acid gives -



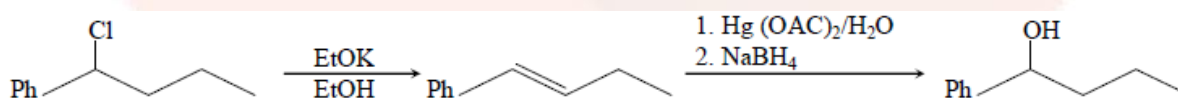
60. Heating of 2-chloro-1-phenylbutane with EtOK/EtOH gives X as the major product. Reaction of X with $\text{Hg}(\text{OAc})_2 / \text{H}_2\text{O}$ followed by NaBH_4 gives Y as the major product, Y is :



Ans: A

Heating of 2-chloro-1-phenyl butane with EtOK/EtOH gives 1-phenyl but-ene (X)

Reaction of X with $\text{Hg}(\text{OAc})_2/\text{H}_2\text{O}$ followed by NaBH_4 gives 1-phenyl butane -1-ol (y)



Mathematics

61. Let S be the set of all $\alpha \in \mathbb{R}$ such that the equation, $\cos 2x + \alpha \sin x = 2\alpha - 7$ has a solution. Then S is equal to :

A. [2, 6] B. [3, 7]

C. [1, 4] D. $\mathbb{R}$

Ans: A

The given trigonometric equation is

$$\cos 2x + \alpha \sin x = 2\alpha - 7$$

$$\Rightarrow 1 - 2\sin^2 x + \alpha \sin x = 2\alpha - 7 \quad [\cos 2x = 1 - 2\sin^2 x]$$

$$\Rightarrow 2\sin^2 x - \alpha \sin x + 2\alpha - 8 = 0$$

$$\Rightarrow \sin x = \frac{\alpha \pm \sqrt{\alpha^2 - 8(2\alpha - 8)}}{4}$$

$$\Rightarrow \sin x = \frac{\alpha \pm (\alpha - 8)}{4}$$

$$\Rightarrow \sin x = \frac{\alpha + \alpha - 8}{4}, \frac{\alpha - \alpha + 8}{4}$$

$$\sin x = 2 \text{ (Not accepted) and } \sin x = \frac{\alpha - 4}{2}$$

Now. As we know that $-1 \leq \sin x \leq 1$

$$-1 \leq \frac{\alpha - 4}{2} \leq 1$$

$$-2 \leq \alpha - 4 \leq 2$$

$$\Rightarrow 2 \leq \alpha \leq 6$$

$$\Rightarrow \alpha \in [2, 6]$$

62. An ellipse, with foci at (0,2) and (0, -2) and minor axis if length 4, passes through which of the following points?

- A. $(\sqrt{2}, 2)$ B. $(1, 2\sqrt{2})$
C. $(2, 2\sqrt{2})$ D. $(2, \sqrt{2})$

Ans: A

Let the equation of ellipse be

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Since, foci at (0,2) and (0, -2) and, major axis is along the y-axis.

Given, $2a = 4$ and $2be = 4$

$$\Rightarrow a = 2, be = 2$$

$$\Rightarrow b^2e^2 = 4$$

$$\Rightarrow b^2 - a^2 = 4$$

$$\Rightarrow b^2 = 8$$

Thus, equation of required ellipse is

$$\frac{x^2}{4} + \frac{y^2}{8} = 1$$

Only option (A) $(\sqrt{2}, 2)$ will satisfy above equation.

63. The derivative of $\tan^{-1}\left(\frac{\sin x - \cos x}{\sin x + \cos x}\right)$, with respect to $\frac{x}{2}$, where

$\left(x \in \left(0, \frac{\pi}{2}\right)\right)$ is:

- A. 2 B. 1
C. $\frac{2}{3}$ D. $\frac{1}{2}$

Ans: A

It is given that Given $y = \tan^{-1} \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right)$

Now, dividing numerator and denominator by $\cos x$, we have,

$$\Rightarrow y = \tan^{-1} \left(\frac{\tan x - 1}{\tan x + 1} \right)$$

$$\Rightarrow y = -\tan^{-1} \left(\frac{1 - \tan x}{1 + \tan x} \right)$$

$$\Rightarrow y = -\tan^{-1} \left[\tan \left(\frac{\pi}{4} - x \right) \right]$$

Since, it is given that $0 < x < \frac{\pi}{2}$, we have,

$$\Rightarrow -\frac{\pi}{2} < -x < 0$$

$$\Rightarrow -\frac{\pi}{4} < \frac{\pi}{4} - x < 0$$

$$\Rightarrow y = -\left(\frac{\pi}{4} - x \right) \quad \left\{ \because \tan^{-1} \tan x = x \forall x \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right) \right\}$$

$$\Rightarrow y = x - \frac{\pi}{4}$$

On derivation y w.r.t. x , we have,

$$\frac{dy}{d(x/2)} = \frac{1}{(1/2)} = 2$$

Thus, the derivative of $\tan^{-1} \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right)$, with respect to $\frac{x}{2}$,

where $\left(x \in \left(0, \frac{\pi}{2} \right) \right)$ is 2.

64. If $a_1, a_2, a_3, \dots$ are in A.P. such that $a_1 + a_7 + a_{16} = 40$, then the sum of the first 15 terms of this A.P. is :

- A. 120 B. 150
C. 200 D. 280

Ans: C

Let the common difference of given AP is 'd'.

$a_1, a_2, a_3, \dots, a_n$ are in A.P.

Since, $a_1 + a_7 + a_{16} = 40$

$$\Rightarrow a + a + 6d + a + 15d = 40$$

$$\Rightarrow 3a + 21d = 40$$

$$\Rightarrow a + 7d = \frac{40}{3} \dots\dots\dots(i)$$

Now sum of first 15 terms is given by

$$S_{15} = \frac{15}{2}[2a + 14d]$$

$$= 15[a + 7d] \dots\dots\dots(ii)$$

From Equation (i) and (ii), we get

$$= 15 \times \frac{40}{3}$$

$$= 200$$

65. Let $\alpha \in (0, \pi/2)$ be fixed. If the integral $\int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx =$

A $(x) \cos 2\alpha + B(x) \sin 2\alpha + C$, where C is a constant of integration, then the functions A(x) and B(x) are respectively :

A. $x + \alpha$ and $\log_e |\sin(x + \alpha)|$

B. $x - \alpha$ and $\log_e |\cos(x - \alpha)|$

C. $x - \alpha$ and $\log_e |\sin(x - \alpha)|$

D. $x + \alpha$ and $\log_e |\sin(x + \alpha)|$

Ans: C

Given that

$$I = \int \frac{\tan x + \tan \alpha}{\tan x - \tan \alpha} dx, \alpha \in (0, \pi/2)$$

$$I = \frac{\frac{\sin x}{\cos x} + \frac{\sin \alpha}{\cos \alpha}}{\frac{\sin x}{\cos x} - \frac{\sin \alpha}{\cos \alpha}} dx$$

$$I = \int \frac{\sin x \cos \alpha + \cos x \sin \alpha}{\sin x \cos \alpha - \cos x \sin \alpha} dx$$

$$I = \int \frac{\sin(x + \alpha)}{\sin(x - \alpha)} dx$$

$$I = \int \frac{\sin(x - \alpha + 2\alpha)}{\sin(x - \alpha)} dx$$

Now, Put $x - \alpha = t \Rightarrow dx = dt$

$$I = \int \frac{\sin(t + 2\alpha)}{\sin t} dt$$

Solving this, we get

$$t(\cos 2\alpha) + (\sin 2\alpha) \log_e |\sin t| + C$$

$$= (x - \alpha) \cos 2\alpha + \sin 2\alpha \log |\sin(x - \alpha)| + C$$

Given that, $A(x) \cos 2\alpha + B(x) \sin 2\alpha + C$, where C is a constant of integration

On comparing we get

$$A(x) = (x - \alpha) \text{ and } B(x) = \log_e |\sin(x - \alpha)|$$

66. A value of $\theta \in (0, \pi/3)$, for which

$$\begin{vmatrix} 1 + \cos^2 \theta & \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & 1 + \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & \sin^2 \theta & 1 + 4 \cos 6\theta \end{vmatrix} = 0, \text{ is :}$$

A. $\frac{7\pi}{36}$

B. $\frac{\pi}{18}$

C. $\frac{\pi}{9}$

D. $\frac{7\pi}{24}$

Ans: C

Given that

$$\theta \in \left(0, \frac{\pi}{3}\right)$$

$$\text{Let } \Delta = \begin{vmatrix} 1 + \cos^2 \theta & \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & 1 + \sin^2 \theta & 4 \cos 6\theta \\ \cos^2 \theta & \sin^2 \theta & 1 + 4 \cos 6\theta \end{vmatrix} = 0$$

Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$, we get

$$\Rightarrow \Delta = \begin{vmatrix} 1 + \cos^2 \theta & \sin^2 \theta & 4 \cos 6\theta \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$$

Applying $C_1 \rightarrow C_1 + C_2$

$$\Rightarrow \Delta = \begin{vmatrix} 2 & \sin^2 \theta & 4 \cos 6\theta \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 2 + 4 \cos 6\theta = 0$$

$$\Rightarrow \cos 6\theta = -\frac{1}{2} = \cos \frac{2\pi}{3}$$

$$\Rightarrow 6\theta = \frac{2\pi}{3} \Rightarrow \theta = \frac{\pi}{9}$$

67. The Boolean expression $\sim(p \Rightarrow (\sim q))$ is equivalent to :

A. $(\sim p) \Rightarrow q$ B. $P \vee q$

C. $Q \Rightarrow \sim p$ D. $P \wedge q$

Ans: D

68. Let $f(x) = 5 - |x - 2|$ and $g(x) = |x + 1|$, $\mathcal{X} \in \mathbf{R}$. If $f(x)$ attains maximum value at α and $g(\mathcal{X})$ attains minimum value at β ,

then $\lim_{x \rightarrow -\alpha\beta} \frac{(x-1)(x^2 - 5x + 6)}{x^2 - 6x + 8}$ is equal to :

A. $3/2$ B. $-1/2$

C. $1/2$ D. $-3/2$

Ans: C

Given function are $f(x) = 5 - |x - 2|$ and $g(x) = |x + 1|$

Clearly, $f(x)$ attains maximum value when

$$|x - 2| = 0 \Rightarrow x = 2 = \alpha$$

And $g(x)$ attains minimum value of $x = -1 = \beta$

$$\Rightarrow \alpha\beta = -2$$

$$\begin{aligned} \lim_{x \rightarrow -\alpha\beta} \frac{(x-1)(x^2-5x+6)}{x^2-6x+8} \\ = \lim_{x \rightarrow 2} \frac{(x-1)(x-2)(x-3)}{(x-2)(x-4)} \quad (\because \alpha\beta = -2) \\ = \frac{(2-1)(2-3)}{(2-4)} = \frac{1}{2} \end{aligned}$$

69. The term independent of x in the expansion of

$$\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6 \text{ is equal to:}$$

A. -72 B. -36

C. -108 D. 36

Ans: B

The given expansion is:

$$\left(\frac{1}{60} - \frac{x^8}{81}\right) \cdot \left(2x^2 - \frac{3}{x^2}\right)^6$$

Now, term independent of x will be $\frac{1}{60} \times$ term independent

$$\text{of } x \text{ in } \left(2x^2 - \frac{3}{x^2}\right)^6 - \frac{1}{8} \times \text{term of } x^{-8} \text{ in } \left(2x^2 - \frac{3}{x^2}\right)^6$$

And

$$\begin{aligned} T_{r+1} \text{ in } \left(2x^2 - \frac{3}{x^2}\right)^6 \text{ will be } T_{r+1} = {}^6C_r (2x^2)^{6-r} \left(-\frac{3}{x^2}\right)^r \\ = {}^6C_r 2^{6-r} (-1)^r \times 3^r \times x^{12-2r-2r} \end{aligned}$$

Now, we have two cases.

Case-I:

For term independent of x is $12 - 4r = 0 \Rightarrow r = 3$

$$T_4 = -{}^6C_3 \times 2^3 \times 3^3 x^6 = -20 \times 2^3 \times 3^3$$

Case-II:

For term of x^{-8} $12 - 4r = -8 \Rightarrow 4r = 20 \Rightarrow r = 5$

$$T_6 = {}^6C_5 \cdot 2^1 \cdot (-1) \cdot 3^5 \cdot x^{-8}$$

Thus, the answer will be:

$$\begin{aligned} &= \frac{1}{60} \times (-20) 2^3 \times 3^3 - \frac{1}{81} \times 6 \times 2 \times (-1) \times 3^5 \\ &= -72 + 36 = -36 \end{aligned}$$

70. Let A , B and C be sets such that $\phi \neq A \cap B \subseteq C$. Then which of the following statements is not true?

A. If $(A - C) \subseteq B$, then $A \subseteq B$

B. $B \cap C \neq \phi$

C. If $(A - B) \subseteq C$, then $A \subseteq C$

D. $(C \cup A) \cap (C \cup B) = C$

Ans: A

Let us consider a sets

$$A = \{a, b, c, d\}$$

$$B = \{c, d, e, f\}$$

$$C = \{a, b, c, d, g, h\}$$

Here,

$$A \cap B = \{c, d\} \subseteq C$$

$$A - C = \phi \not\subseteq B$$

But $A \not\subset B$

71. If $[X]$ denotes the greatest integer $\leq X$, then the system of linear equations

$$[\sin \theta]x + [-\cos \theta]y = 0$$

A. has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and have infinitely many solutions if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

B. have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and has a unique solution if $\theta \in \left(\pi, \frac{7\pi}{6}\right)$

C. have infinitely many solutions if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$.

D. has a unique solution if $\theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \cup \left(\pi, \frac{7\pi}{6}\right)$

Ans: B

Given system of linear equations is

$$[\sin \theta]x + [-\cos \theta]y = 0 \quad \dots (i)$$

$$\text{And } [\cot \theta]x + y = 0 \quad \dots (ii)$$

Where, $[x]$ denotes the greatest integer function $\leq x$

There are two cases.

Case-I :

$$\text{When } \theta \in \left(\frac{\pi}{2}, \frac{2\pi}{3}\right) \text{ and } \sin \theta \in \left(\frac{\sqrt{3}}{2}, 1\right)$$

$$[\sin \theta] = 0$$

$$\cos \theta \in \left(-\frac{1}{2}, 0\right) \text{ and } -\cos \theta \in \left(0, \frac{1}{2}\right)$$

$$[-\cos \theta] = 0$$

$$\text{and } \cot \theta \in \left(\frac{-1}{\sqrt{3}}, 0\right)$$

$$[\cot \theta] = -1$$

$$[\sin \theta] = 0, [-\cos \theta] = 0, [\cot \theta] = -1$$

equation (i) and (ii) will

$$\begin{cases} 0x + 0y = 0 \\ -x + y = 0 \end{cases} \text{ system will have infinitely many solution}$$

Case-II:

$$\text{When } \theta \in \left(\pi, \frac{7\pi}{6}\right) \text{ and } \sin \theta \in \left(-\frac{1}{2}, 0\right)$$

$$[\sin \theta] = -1$$

$$\cos \theta \in \left(-1, \frac{-\sqrt{3}}{2}\right)$$

$$[\cos \theta] = 1$$

$$\cot \theta \in (\sqrt{3}, \infty) \Rightarrow [\cot \theta] = n$$

$$[\cot \theta] = \{1, 2, 3, \dots\}$$

$$[\sin \theta] = -1, [\cos \theta] = 1$$

$$[\cot \theta] = \{1, 2, 3, \dots\}$$

equation (i) and (ii) will

$$-x - y = 0$$

$$Ix + y = 0 \quad I = \{1, 2, \dots\}$$

It will have unique solution is all cases $x = 0, y = 0$

72. The equation of a common tangent to the curves, $y^2 = 16x$ and $xy = -4$, is:

- A. $2x - y + 2 = 0$ B. $x - 2y + 16 = 0$
C. $x - y + 4 = 0$ D. $2 - y + 4 = 0$

Ans: C

Given equation of curves are

$y^2 = 16x$ (parabola)

And $xy = -4$ (Rectangular hyperbola)

$y = mx + \frac{4}{m}$ is always tangent to $y^2 = 16x$... (i)

if it is tangent to the $xy = -4$

$$x\left(mx + \frac{4}{m}\right) = -4$$

$$m^2x^2 + 4x = -4m$$

$$m^2x^2 + 4x + 4m = 0$$

for tangent $D = 0$

$$16 - 16m^3 = 0 \Rightarrow m = 1$$

put the value of m in equation (i)

$$y = x + 4$$

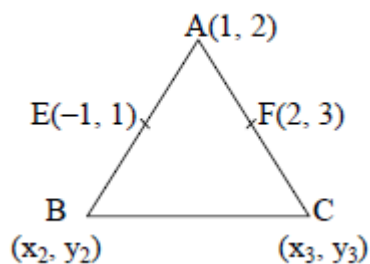
Equation of common tangent is $x - y + 4 = 0$

73. A triangle has a vertex at (1,2) and the mid points of the two sides through it are (-1,1) and (2,3). Then the centroid of this triangle is :

- A. $\left(1, \frac{7}{3}\right)$ B. $\left(\frac{1}{3}, \frac{5}{3}\right)$
C. $\left(\frac{1}{3}, 2\right)$ D. $\left(\frac{1}{3}, 1\right)$

Ans: C

Let a $\triangle ABC$ is such that vertices $A(1,2)$, $B(x_1, y_1)$ and $C(x_2, y_2)$



It is given that mid points of side AB is $(-1, 1)$

$$\frac{x_2 + 1}{2} = -1, \quad \frac{y_2 + 2}{2} = 1$$

$$x_2 = -3, y_2 = 0$$

So, point B is $(-3, 0)$

It is given that mid points of side AC is $(2, 3)$

$$\frac{x_3 + 1}{2} = 2 \quad \& \quad \frac{y_3 + 2}{2} = 3$$

$$x_3 = 3, y_3 = 4$$

So, point C is $(3, 4)$

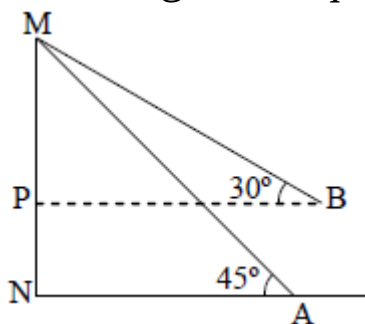
$$\text{now, centroid of } \triangle ABC = \left(\frac{1 + (-3) + 3}{3}, \frac{2 + 0 + 4}{3} \right) = G \left(\frac{1}{3}, 2 \right)$$

74. The angle of elevation of the top of a vertical tower standing on a horizontal plane is observed to be 45° from a point A on the plane. Let B be the point 30 m vertically above the point A. If the angle of elevation of the top of the tower from B be 30° , then the distance (in m) of the foot of the tower from the point A is :

- | | |
|-----------------------|-----------------------|
| A. $15(3 - \sqrt{3})$ | B. $15(1 + \sqrt{3})$ |
| C. $15(3 + \sqrt{3})$ | D. $15(5 - \sqrt{3})$ |

Ans: C

According to the question, we have the following figure.



$$AB = 30 \text{ m} = NP$$

In $\triangle ANM$

$$\tan 45^\circ = \frac{MN}{AN} = 1$$

$$\Rightarrow MN = AN$$

$$PM = MN - 30$$

$$= AN - 30$$

In $\triangle BPM$

$$\tan 30^\circ = \frac{PM}{PB} = \frac{AN - 30}{AN}$$

$$\frac{1}{\sqrt{3}} = \frac{AN - 30}{AN}$$

$$AN = \sqrt{3} AN - 30\sqrt{3}$$

$$AN = \frac{30\sqrt{3}}{\sqrt{3} - 1} = \frac{30\sqrt{3}(\sqrt{3} + 1)}{2} = 15(3 + \sqrt{3})$$

Therefore, the distance (in m) of the foot of the tower from the point A is $15(3 + \sqrt{3})$

75. Let $\alpha \in \mathbf{R}$ and the three vectors $\vec{a} = \alpha\hat{i} + \hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} + \hat{j} - \alpha\hat{k}$ and $\vec{c} = \alpha\hat{i} + 2\hat{j} + 3\hat{k}$. Then the set $S = \{\alpha : \vec{a}, \vec{b} \text{ and } \vec{c} \text{ are coplanar}\}$

A. Contains exactly two numbers only one of which is positive

B. Is empty

C. Is singleton

D. Contains exactly two positive numbers

Ans: B

If $\vec{a}, \vec{b}, \vec{c}$ are coplanar, then

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0$$

$$\begin{vmatrix} \alpha & 1 & 3 \\ 2 & 1 & -\alpha \\ \alpha & -2 & 3 \end{vmatrix} = 0$$

$$\alpha(3 - 2\alpha) + 1(-\alpha^2 - 6) + 3(-4 - \alpha) = 0$$

$$3\alpha - 2\alpha^2 - \alpha^2 - 6 - 12 - 3\alpha = 0$$

$$-3\alpha^2 - 18 = 0$$

$$\alpha^2 + 6 = 0$$

there is no value of α for which $\alpha^2 + 6$ becomes zero.

So, $[\vec{a} \ \vec{b} \ \vec{c}] \neq 0$ and $\vec{a}, \vec{b}, \vec{c}$ are not coplanar.

Thus, S is empty set

76. A value of α such that $\int_{\alpha}^{\alpha+1} \frac{dx}{(x + \alpha)(x + \alpha + 1)} = \log_e \left(\frac{9}{8} \right)$ is:

A. $\frac{1}{2}$ B. -2

C. $-\frac{1}{2}$ D. 2

Ans: B

Given that

$$\begin{aligned} \int_{\alpha}^{\alpha+1} \frac{dx}{(x+\alpha)(x+\alpha+1)} &= \log_e \left(\frac{9}{8} \right) \\ \Rightarrow \int_{\alpha}^{\alpha+1} \frac{(x+\alpha+1) - (x+\alpha)}{(x+\alpha)(x+\alpha+1)} dx &= \log_e \left(\frac{9}{8} \right) \\ \Rightarrow \int_{\alpha}^{\alpha+1} \frac{dx}{x+\alpha} - \int_{\alpha}^{\alpha+1} \frac{dx}{x+\alpha+1} &= \log_e \left(\frac{9}{8} \right) \\ \Rightarrow \log_e \left(\frac{x+\alpha}{x+\alpha+1} \right) \Big|_{\alpha}^{\alpha+1} &= \log_e \left(\frac{9}{8} \right) \\ \Rightarrow \log_e \left(\frac{2\alpha+1}{2\alpha+2} \right) - \log_e \left(\frac{2\alpha}{2\alpha+1} \right) &= \log_e \left(\frac{9}{8} \right) \\ \Rightarrow \log \left[\left(\frac{2\alpha+1}{2\alpha+2} \right) \left(\frac{2\alpha+1}{2\alpha} \right) \right] &= \log_e \frac{9}{8} \\ \Rightarrow \frac{(2\alpha+1)^2}{4\alpha(\alpha+1)} &= \frac{9}{8} \\ \Rightarrow 8\alpha^2 + 8\alpha + 2 &= 9\alpha^2 + 9\alpha \\ \Rightarrow \alpha^2 + \alpha - 2 &= 0 \\ \Rightarrow (\alpha+2)(\alpha-1) &= 0 \\ \Rightarrow \alpha &= 1, -2 \end{aligned}$$

From the options, we get $\alpha = -2$

77. A plane which bisects the angle between the two given planes $2X - Y + 2z - 4 = 0$ and $X + 2Y + 2z - 2 = 0$, passes through the point :

- A. (1, 4, -1) B. (2, -4, 1)
C. (1, -4, 1) D. (2, 4, 1)

Ans: B

Equation of given planes are

$$2X - Y + 2z - 4 = 0 \dots\dots\dots(i)$$

$$\text{And } X + 2Y + 2z - 2 = 0 \dots\dots\dots(ii)$$

Now, equation of planes bisecting the angles between planes (i) and (ii) are

$$\frac{2x - y + 2z - 4}{\sqrt{2^2 + (-1)^2 + 2^2}} = \pm \left(\frac{x + 2y + 2z - 2}{\sqrt{1^2 + 2^2 + 2^2}} \right) \dots (iii)$$

Case I : take positive sign

$$2x - y + 2z - 4 = x + 2y + 2z - 2$$

$$x - 3y - 2 = 0 \dots (iv)$$

Case-II : take negative sign

$$2x - y + 2z - 4 = -(x + 2y + 2z - 2)$$

$$2x - y + 2z - 4 = -x - 2y - 2z + 2$$

$$3x + y + 4z - 6 = 0 \dots (v)$$

Now from given options, Point (2, -4, 1) satisfy equation (v).

78. The length of the perpendicular drawn from the point (2, 1, 4) to the plane containing the lines $\vec{r} = (\hat{i} + \hat{j}) + \lambda(-\hat{i} + \hat{j} - 2\hat{k})$ and $\vec{r} = (\hat{i} + \hat{j}) + \mu(-\hat{i} + \hat{j} - 2\hat{k})$ is :

A. $\frac{1}{3}$

B. $\frac{1}{\sqrt{3}}$

C. 3

D. $\sqrt{3}$

Ans: D

Given that

$$\vec{r} = (\hat{i} + \hat{j}) + \lambda(-\hat{i} + \hat{j} - 2\hat{k})$$

And

$$\vec{r} = (\hat{i} + \hat{j}) + \mu(-\hat{i} + \hat{j} - 2\hat{k})$$

As we know that equation of plane containing two given lines is

$$\begin{vmatrix} x-1 & y-1 & z \\ 1 & 2 & -1 \\ -1 & 1 & -2 \end{vmatrix} = 0$$

$$(x-1)(-4+1) + (y-1)(1+2) + z(1+2) = 0$$

$$-3(x-1) + 3(y-1) + 3z = 0$$

$$x - y - z = 0$$

The length of perpendicular from (2, 1, 4) to the plane $x - y - z = 0$ is given by

$$\left| \frac{2-1-4}{\sqrt{1^2+1^2+1^2}} \right| = \frac{3}{\sqrt{3}} = \sqrt{3}$$

79. $\lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}}$ is :

A. 3 B. 1

C. 2 D. 6

Ans: C

Let $P = \lim_{x \rightarrow 0} \frac{x + 2 \sin x}{\sqrt{x^2 + 2 \sin x + 1} - \sqrt{\sin^2 x - x + 1}}$ $\left(\frac{0}{0} \text{ form} \right)$

On rationalization, we get

$$P = \lim_{x \rightarrow 0} \frac{x + 2 \sin x}{x^2 + 2 \sin x + 1 - \sin^2 x + x - 1} \times \left(\sqrt{x^2 + 2 \sin x + 1} + \sqrt{\sin^2 x - x + 1} \right)$$

$$P = \lim_{x \rightarrow 0} \frac{x + 2 \sin x}{x^2 + 2 \sin x - \sin^2 x + x} \times (2) \left(\frac{0}{0} \text{ form} \right)$$

Now, applying L'Hopital's Rule, we get

$$P = \lim_{x \rightarrow 0} \frac{2(1 + 2 \cos x)}{2x + 2 \cos x - 2 \sin x \cos x + 1}$$

$$P = \frac{2(3)}{2+1} = 2 \quad (\text{on applying limit})$$

80. A circle touching the x-axis at (3, 0) and making an intercept of length 8 on the y-axis passes through the point :

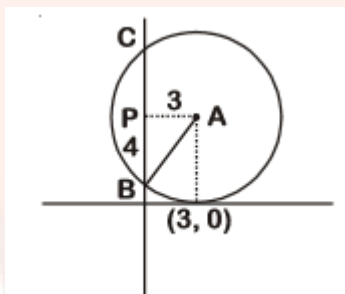
A. (2,3) B. (1,5)

C. (3,5)

D. (3, 10)

Ans: D

It is given that the circle touching the x-axis at (3, 0) and making an intercept of length 8 on the y-axis. Let centre of circle is A and circle cuts the y axis at B and C. Let midpoint of chord BC is P.



$AB = 5 = \text{Radius of circle}$

Now the equation of circle having centre (3, ± 5) and radius = 5 is

$$(x-3)^2 + (y \pm 5)^2 = 25$$

Equation of circle is : $(x-3)^2 + (y-5)^2 = 25$

Now, from the options Only (3, 10) satisfies this equation.

81. A person throws two fair dice. He wins Rs. 15 for throwing a doublet (same numbers on the two dice), wins Rs. 12 when the throw results in the sum of 9, and loses Rs. 6 for any other outcome on the throw. Then the expected gain/loss (in Rs.) of the person is :

A. $\frac{1}{2}$ gain

B. $\frac{1}{4}$ loss

C. 2 gain

D. $\frac{1}{2}$ loss

Ans: D

It is given that a person wins

Rs. 15 for throwing a doublet

Favourable cases = (1,1) (2,2) (3,3) (4,4) (5,5) (6,6) and Total cases = 36

$$\text{Probability of doublet} = \frac{6}{36}$$

Rs. 12 when the throw results in the sum of 9

Favourable cases = (6,3) (4,5) (5,4) (3,6) and Total cases = 36

$$\text{Probability of sum of 9} = \frac{4}{36}$$

$$\text{Other probability} = \frac{26}{36}$$

Win	+15	+12	-6
Prob.	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{26}{36}$

$$\text{Expected gain/loss} = 15 \times \frac{6}{36} + 12 \times \frac{4}{36} - 6 \times \frac{26}{36}$$

$$= \frac{90}{36} + \frac{48}{36} - \frac{156}{36} = \frac{-1}{2} \Rightarrow \frac{-1}{2}$$

Its means loss of Rs $\frac{1}{2}$

82. Let $z \in \mathbb{C}$ with $\text{Im}(z) = 10$ and it satisfies $\frac{2z - n}{2z + n} = 2i - 1$ for some

natural number n. Then :

A. $n = 40$ and $\text{Re}(z) = -10$

B. $n = 20$ and $\text{Re}(z) = -10$

C. $n = 40$ and $\text{Re}(z) = 10$

D. $n = 20$ and $\text{Re}(z) = 10$

Ans: B

Let $z = x + 10i$, as $\text{Im}(z) = 10$ (given)

Since, z Satisfies

$$\text{Given } \frac{2z - n}{2z + n} = 2i - 1$$

$$\Rightarrow \frac{2(x + 10i) - n}{2(x + 10i) + n} = 2i - 1$$

$$\Rightarrow (2x - n) + 20i = (2i - 1) [(2x + n) + 20i]$$

Comparing real and imaginary part

$$\Rightarrow 2x - n = 2(-20) - (2x + n) \text{ and } 20 = 2(2x + n) - 20$$

$$\Rightarrow 2x - n = -40 - 2x - n \text{ and } 20 = 4x + 2n - 20$$

$$\Rightarrow 4x = -40 \text{ and } 4x + 2n = 40$$

$$\Rightarrow x = -10 \text{ and } -40 + 2n = 40$$

$$\Rightarrow n = + 40$$

$$\Rightarrow n = 40 \text{ and } \text{Re}(z) = -10$$

83. for an initial screening of an admission test, a candidate is given fifty problems to solve. If the probability that the candidate can solve any problem is $\frac{4}{5}$, then the probability that he is unable to solve less than two problems is :

A. $\frac{54}{5} \left(\frac{4}{5}\right)^{49}$

B. $\frac{201}{5} \left(\frac{1}{5}\right)^{49}$

C. $\frac{316}{25} \left(\frac{4}{5}\right)^{48}$

D. $\frac{164}{25} \left(\frac{1}{5}\right)^{48}$

Ans: A

Given that

The probability that the candidate can solve any problem = $\frac{4}{5}$

The probability that the candidate can solve any problem

$$= 1 - \frac{4}{5} = \frac{1}{5}$$

Total problems = 50

P(unable to solve less than two problems)

= P(not solving one problem) + P(not solving zero problem)

$$= {}^{50}C_0 \left(\frac{1}{5}\right)^0 \left(\frac{4}{5}\right)^{50} + {}^{50}C_1 \left(\frac{1}{5}\right)^1 \left(\frac{4}{5}\right)^{49}$$

$$= \frac{4^{50}}{5^{50}} + 50 \cdot \frac{4^{49}}{5 \cdot 5^{49}}$$

$$= \left(\frac{4}{5}\right)^{49} + \left(\frac{4}{5} + 10\right)$$

$$= \frac{54}{5} \cdot \left(\frac{4}{5}\right)^{49}$$

84. If α , β and γ are three consecutive terms of a non-constant G.P. such that the equations $\alpha x^2 + 2\beta x + \gamma = 0$ and $x^2 + x - 1 = 0$ have a common root, the $(\beta + \gamma)$ is equal to :

- A. $\alpha\beta$ B. 0
C. $\alpha\gamma$ D. $\beta\gamma$

Ans: D

Given, α , β and γ are three consecutive terms of a non-constant G.P.

$\alpha x^2 + 2\beta x + \gamma = 0$ & $x^2 + x - 1 = 0$ have a common roots. Both roots will be common

$$\frac{\alpha}{1} = \frac{2\beta}{1} = \frac{\gamma}{-1} = \lambda$$

$$\alpha = \lambda, \beta = \frac{\lambda}{2}, \gamma = -\lambda$$

Now,

$$\alpha(\beta + \gamma) = \lambda \left(\frac{\lambda}{2} - \lambda \right) = \frac{-\lambda^2}{2} = \beta\gamma$$

85. If the area (in sq. units) bounded by the parabola $y^2 = 4\lambda x$ and the line $y = \lambda x$, $\lambda > 0$, is $\frac{1}{9}$, then λ is equal to :

- A. 48 B. $4\sqrt{3}$
C. $2\sqrt{6}$ D. 24

Ans: D

Given, equation of curves are

$$y^2 = 4\lambda x \text{ and } y = \lambda x$$

the intersection point will get on the solving equation (i) and (ii), we get

$$\lambda^2 x^2 = 4\lambda x$$

$$x = 0, x = \frac{4}{\lambda}$$

$$\text{Required area} = \int_0^{4/\lambda} (\sqrt{4\lambda x} - \lambda x) dx = \frac{1}{9} \text{ (Given)}$$

$$= 2\sqrt{\lambda} = \left[\frac{x^{3/2}}{\frac{3}{2}} \right]_0^{4/\lambda} - \lambda \left[\frac{x^2}{2} \right]_0^{4/\lambda} = \frac{1}{9}$$

$$\frac{4}{3} \sqrt{\lambda} \times \frac{(2^2)^{3/2} x}{\lambda^{3/2}} - \frac{\lambda}{2} \times \frac{16}{\lambda^2} = \frac{1}{9}$$

$$\Rightarrow \frac{8}{3\lambda} = \frac{1}{9} \Rightarrow \lambda = 24$$

86. A group of students comprises of 5 boys and n girls. If the number of ways, in which a team of 3 students can randomly be selected from this group such that there is at least one boy and at least one girl in each team, is 1750, then n is equal to :

- A. 27 B. 24
C. 28 D. 25

Ans: D

Given that a group of students comprises of 5 boys and n girls.

Number of ways of selecting three persons such that there is atleast one boy and atleast one girl in the selected persons =

$${}^5C_1 \cdot {}^nC_2 + {}^5C_2 \cdot {}^nC_1$$

$$\Rightarrow {}^5C_1 \cdot {}^nC_2 + {}^5C_2 \cdot {}^nC_1 = 1750 \quad (\text{Given})$$

$$\Rightarrow \frac{5n(n-1)}{2} + 10n = 1750$$

$$\Rightarrow \frac{n(n-1)}{2} + 2n = 350$$

$$\Rightarrow n^2 + 3n = 700$$

$$\Rightarrow n^2 + 3n - 700 = 0$$

$$\Rightarrow n = 25$$

87. The general solution of the differential equation $(y^2 - x^3) dx - xy dy = 0$ ($x \neq 0$) is : (where c is a constant of integration)

- A. $y^2 - 2x^3 + cx^2 = 0$ B. $y^2 + 2x^2 + cx^3 = 0$
C. $y^2 - 2x^2 + cx^3 = 0$ D. $y^2 + 2x^3 + cx^2 = 0$

Ans: D

Given that differential equation is

$$(y^2 - x^3)dx - xy dy = 0 \quad (x \neq 0)$$

$$\Rightarrow y^2 - x^3 - xy \frac{dy}{dx} = 0$$

$$\Rightarrow xy \frac{dy}{dx} - y^2 = -x^3$$

$$\Rightarrow y \frac{dy}{dx} - \frac{1}{x} y^2 = -x^2 \quad \dots(i)$$

$$\text{Let } y^2 = v \Rightarrow 2y \frac{dy}{dx} = \frac{dv}{dx}$$

put in equation (i)

$$\frac{1}{2} \frac{dv}{dx} - \frac{1}{x} v = -x^2$$

$$\Rightarrow \frac{dv}{dx} + \left(-\frac{2}{x}\right)v = -2x^2 \quad \dots(ii)$$

Which is the linear differential equation.

Now

$$\text{I.F.} = e^{\int -\frac{2}{x} dx} = e^{-2 \ln x} = \frac{1}{x^2}$$

$\therefore$ solution of linear differential equation is given by

$$v \times \frac{1}{x^2} = \int -2x^2 \times \frac{1}{x^2} dx - C$$

$$\Rightarrow \frac{v}{x^2} = -2x - C$$

$$\Rightarrow y^2 = -2x^3 - cx^2$$

$$\Rightarrow y^2 + 2x^3 + cx^2 = 0$$

88. A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$. Then an equation of the line L is:

A. $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$

B. $\sqrt{3}x + y = 8$

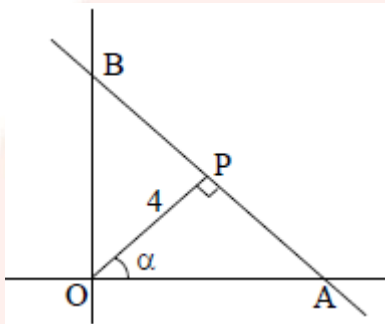
C. $x + \sqrt{3}y = 8$

D. $(\sqrt{3} - 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$

Ans: A, D

Given that a straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$.

According to the question, we have following figure.



$OP = 4$

Given OP makes 60° with $x + y = 0$

Let slope of OP = m

$$\Rightarrow \tan 60^\circ = \left| \frac{m+1}{1-m} \right|$$

$$\Rightarrow \frac{m+1}{m-1} = \sqrt{3} \text{ or } -\sqrt{3}$$

$$\Rightarrow m(\sqrt{3} - 1) = \sqrt{3} + 1 \text{ or } m(1 + \sqrt{3}) = \sqrt{3} - 1$$

$$\Rightarrow m = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \text{ or } m = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$\Rightarrow \tan \alpha = \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \text{ or } \tan \alpha = \frac{\sqrt{3} - 1}{\sqrt{3} + 1}$$

$$\Rightarrow \tan \alpha = \tan 75^\circ \text{ or } \tan \alpha = \tan 15^\circ$$

$$\Rightarrow \alpha = 75^\circ \text{ or } \alpha = 15^\circ$$

Then the perpendicular makes an angle of 15° or 75° with x-axis.

So the equation of line will be

$$x \cos 75^\circ + y \sin 75^\circ = 4 \text{ or}$$

$$x \cos 15^\circ + y \sin 15^\circ = 4 \quad (\because x \cos \alpha + y \sin \alpha = P)$$

$$\Rightarrow (\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$$

$$\text{or } (\sqrt{3} - 1)x + (\sqrt{3} + 1)y = 8\sqrt{2}$$

89. The tangents to the curve $y = (x - 2)^2 - 1$ at its points of intersection with the line $x - y = 3$, intersect at the point:

A. $\left(\frac{5}{2}, 1\right)$ B. $\left(\frac{5}{2}, -1\right)$

C. $\left(-\frac{5}{2}, 1\right)$ D. $\left(-\frac{5}{2}, -1\right)$

Ans: B

Given equation of parabola is

$$y = (x - 2)^2 - 1$$

$$\Rightarrow y = x^2 - 4x + 3$$

$x - y - 3 = 0 \dots$ (i) will be chord of contact of parabola

$$y = x^2 - 4x + 3$$

Let the required point is $P(x_1, y_1)$ chord of contact for point P is

$$\frac{y + y_1}{2} = xx_1 - 4 \frac{(x + x_1)}{2} + 3$$

$$y + y_1 = 2x_1x - 4x - 4x_1 + 6$$

$$(2x_1 - 4)x - y + (-4x_1 - y_1 + 6) = 0 \dots \text{(ii)}$$

Equation (i) & (ii) are same line

$$\frac{2x_1 - 4}{1} = \frac{-1}{-1} = \frac{-4x_1 - y_1 + 6}{-3}$$

$$\Rightarrow 2x_1 - 4 = 1$$

$$x_1 = \frac{5}{2}$$

$$-4x_1 - y_1 + 6 = -3$$

$$-10 = y_1 + 9 = 0$$

So, the required point is $\left(\frac{5}{2}, -1\right)$

90. If ${}^{20}C_1 + (2^2){}^{20}C_2 + (3^2){}^{20}C_3 + \dots + (20^2){}^{20}C_{20} = A(2^B)$, then the ordered pair (A, B) is equal to :

A. (380, 18) B. (420, 18)

C. (380, 19) D. (420, 19)

Ans: B

As we know that

$$(1 + x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$$

For $n = 20$, we get

$$(1 + x)^{20} = {}^{20}C_0 + {}^{20}C_1x + {}^{20}C_2x^2 + \dots + {}^{20}C_{20}x^{20} \dots (i)$$

On differentiating both sides of eqn (i) w.r.t. x, we get

$$20(1 + x)^{19} = {}^{20}C_1 \cdot 1 + 2 \cdot {}^{20}C_2x + \dots + 20 \cdot {}^{20}C_{20}x^{19} \dots (ii)$$

On multiply both sides of eqn (ii) by x, we get

$$20x(1 + x)^{19} = {}^{20}C_1 \cdot x + 2 \cdot {}^{20}C_2x^2 + \dots + 20 \cdot {}^{20}C_{20}x^{20} \dots (iii)$$

Again differentiating both sides of eqn (iii) w.r.t. x, we get

$$20[(1 + x)^{19} + 19x(1 + x)^{18}] = 1 \cdot {}^{20}C_1 + 2^2 \cdot {}^{20}C_2x + \dots + (20^2) \cdot {}^{20}C_{20}x^{19} \dots (iv)$$

Putting value of $x = 1$ in eqn (iv)

$$\begin{aligned} 20(2^{19} + 19 \cdot 2^{18}) &= 1^2 \cdot {}^{20}C_1 + 2^2 \cdot {}^{20}C_2 + \dots + (20^2) \cdot {}^{20}C_{20} \\ &= 20 \times 2^{18}(2 + 19) = 20 \times 21 \times 2^{18} = 420 \times 2^{18} \\ &= A(2^B) \dots (v) \end{aligned}$$

On comparing, we get

$$(A, B) = (420, 18)$$

